

A Literature Review on

”Finance and Development: A Tale of Two Sectors”

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Contents

1	Introduction	2
2	The model	2
2.1	Model Setup	2
2.1.1	The Utility Function	3
2.1.2	Production Functions	4
2.1.3	Investment and Capital Choice	5
2.1.4	Credit and Rental Markets	6
2.1.5	The Case of Perfect-Credit Market	8
2.2	Individual's Optimization Problem	12
3	The Market Equilibrium	14
4	Implementation - Decentralized Model with Occupational Choice	16
5	Results	17
5.1	Baseline policy functions of occupational choice and consumption	17
5.2	Change in technological parameters	19
5.2.1	Changes in real wages	19
5.2.2	Changes in interest rate	22
5.2.3	Changes in depreciation rate	26
5.2.4	Conclusion on the effect of technological parameters	29
5.3	Change in Entry/Exit Criterion on Self-Employed	30
5.4	Change in Capital Rental Limit	32
6	Contract Enforcement on Policy Functions in General Equilibrium	34
	Appendices	36
A	Baseline Policy Functions	36
A.1	Saving policy functions	36
B	Sensitivity of Policy Functions when changing technological parameters	36
B.1	Other Policy functions at different values of real wages	36
B.2	Other Policy functions at different values of depreciation rates	37

Dynamic Macroeconomics

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A LITERATURE REVIEW ON
"Finance and Development: A Tale of Two Sectors"

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1. Introduction

The seminal work by Buera et al. (2011) provides a quantitative framework to analyze the impact of financial frictions on aggregate total factor productivity (TFP), sector-level productivity differences, and capital misallocation. Their framework highlights how imperfect financial markets disproportionately affect manufacturing relative to services, leading to lower aggregate output and productivity in financially underdeveloped economies.

The following sections of the report will be organized as follows: We begin by formally defining the baseline model (section 2), which outlines household preferences, production technologies, capital investment, and credit market frictions, and the occupational decision framework. Section 2.2 also includes individual optimization behavior and characterizes decision rules under borrowing constraints and rental market imperfections (section 2.1.4). Building on this, Section 3 describes the general equilibrium conditions. Section 4 implements the decentralized version of the model, incorporating heterogeneous occupational choices. Section 5 presents the main quantitative results. We begin with the baseline policy functions that govern occupational choice and consumption behavior. Then we explore how changes in technological parameters—specifically the real wage, interest rate, depreciation rate, fixed costs, and entry/ exit barrier—affect household behavior and equilibrium outcomes. Finally, Section 6 evaluates the implications of contract enforcement mechanisms for general equilibrium outcomes.

2. The model

2.1. Model Setup

Buera et al. (2011) construct an economy with two distinct sectors: services (**S**), a **small-scale** sector where output is used exclusively for consumption, and manufacturing (**M**), a **large-scale** sector where output serves both consumption and investment purposes. Manufactured goods act as the numeraire in the economy.

Population and Heterogeneity

The economy consists of a measure N of infinitely lived individuals, each heterogeneous in wealth (a), which evolves endogenously through savings and investment, and entrepreneurial talent ($\mathbf{z} = (z_S, z_M)$), which determines their productivity in either sector.

Entrepreneurial Talent Dynamics

Entrepreneurial ability follows a stochastic process: with persistence γ indicating the probability of retaining entrepreneurial ability, while with probability $1 - \gamma$, their ability is reset, and a new talent vector is drawn from a distribution $\mu(\mathbf{z})$.

Occupational Choice

Each period, individuals choose whether to work for a wage or operate a business in one of the two sectors j (where $j = S, M$). Entrepreneurs must pay a sector-specific fixed cost κ_j , where $\kappa_M > \kappa_S$, making manufacturing more capital-intensive than services.

Market Frictions and Capital Allocation

A crucial assumption in the model is that capital markets are imperfect, meaning access to credit is limited by collateral constraints, which restrict borrowing based on individual wealth. Additionally, entrepreneurial talent is inalienable, meaning there is no market for managers; each entrepreneur must operate their own firm. This assumption follows the span-of-control framework by Lucas (1978) and extensions by Rossi-Hansberg and Wright (2007), where managerial talent is a key determinant of firm size.

In this economy, the model incorporates three key players: individuals, who act as both consumers (section 2.1.1) and producers (as they are assumed to be entrepreneurs) (section 2.1.2), and the credit market, which ensures a comprehensive framework (section 2.1.4).

2.1.1. The Utility Function

To analyze how individuals allocate consumption over time and across different goods, Buera et al. (2011) defines their preferences using an **expected lifetime utility function**. Their objective is to maximize expected lifetime utility by choosing consumption paths over time while considering the persistence of their entrepreneurial talent and income-generating opportunities. This function captures two key dimensions of decision-making: (1) **Intertemporal** consumption choices: Individuals must decide how to allocate resources over time based on their discount factor β . A higher β indicates that individuals value future consumption more highly relative to present consumption; and (2) **Intratemporal** consumption choices, where individuals also decide how to distribute consumption between services c_S and manufactured goods c_M substitute for each other in each period.

The expected utility function reflects constant relative risk aversion (CRRA) preferences:

$$U(c) = \mathbb{E} \left[\sum_{t=0}^{\infty} \beta^t u(c_t) \right]$$

where the per-period utility function is given by:

$$u(c_t) = \frac{1}{1-\sigma} \left(\psi c_{S,t}^{1-1/\varepsilon} + (1-\psi) c_{M,t}^{1-1/\varepsilon} \right)^{(1-\sigma)/(1-1/\varepsilon)} \quad (1)$$

where the coefficient of risk aversion σ also serves as the inverse of the elasticity of intertemporal substitution (EIS)¹, ε is the intratemporal elasticity of substitution that determines how easily individuals can switch between consuming services and manufactured goods², and parameter ψ represents

¹The EIS σ controls risk aversion. If σ is high, individuals strongly prefer smooth consumption paths and are less responsive to interest rate changes. Vice versa, when σ is low, individuals are more flexible in adjusting consumption over time.

²When $\varepsilon > 1$, services and manufactured goods are easily substitutable, i.e., a price increase in one leads to higher consumption of the other. If $\varepsilon < 1$, the two goods are complements, meaning consumers prefer to consume them in fixed proportions.

the expenditure share on services, influencing the relative weight of services in consumption decisions ³.

The structure represents trade-offs and distortions when individuals optimize their financial decision-making: When financial markets are imperfect, individuals may struggle to maintain smooth consumption over time (σ effect) or may be unable to adjust consumption optimally between sectors (ε effect).

2.1.2. Production Functions

Firstly, we assume that households are born as entrepreneurs in one of two sectors: manufacturing (**M**) or services (**S**), as mentioned in 2.1 ⁴. At the beginning of each period, entrepreneurs must pay a sector-specific fixed cost κ_j (where $j \in M, S$) to operate their businesses. These costs are denominated in the output of their respective sector. A key assumption is that the fixed cost for operating in the manufacturing sector is higher than that in the service sector, i.e., $\kappa_M > \kappa_S$. ⁵ These costs are incurred each period, meaning they represent ongoing operational expenses rather than one-time setup costs. After paying the fixed cost, an entrepreneur with talent z_j in sector j produces output using capital k and labor l according to

$$z_j f(k, l) = z_j k^\alpha l^\theta \quad (2)$$

from (2), we have the **sector-specific** production functions as follows:

Production in the Manufacturing Sector

$$y_M(k, l; z_M) = z_M k_M^\alpha l_M^\theta \quad (3)$$

Production in the Service Sector

$$y_S(k, l; z_S) = z_S k_S^\alpha l_S^\theta \quad (4)$$

where α_j and θ_j are the elasticities of output with respect to capital and labor, respectively. We assume diminishing returns to scale at the establishment level, meaning:

$$\alpha_j + \theta_j < 1, \quad \text{for } j \in \{M, S\}$$

Buera et al. (2011) assumes that factor elasticities are identical across sectors (as according to Chari et al. (1996); Ákos Valentinyi and Herrendorf (2008)).

Given factor prices w (wage) and R (rental rate of capital), the general entrepreneur's profit function is given by:

$$\pi_j(k, l; R, w, \mathbf{p}) = p_j z_j k^\alpha l^\theta - Rk - wl - (1 + r)p_j \kappa_j \quad (5)$$

in which, in each sector:

³The parameter ψ determines the relative importance of services in total consumption. A higher ψ means individuals allocate more spending to services, making them more dependent on the service sector, and vice versa, a lower ψ implies a stronger reliance on manufactured goods.

⁴This assumption is an extension as according to the requirement of the assignment. This distinction implies that the economy consists of **two types of consumption goods**, **two distinct production functions** with **different productivity processes**, and **two separate investment equations** governing capital accumulation in each sector.

⁵Buera et al. (2011) adopts this assumption from their empirical evidence in their work, stating that the large-scale manufacturing sector has higher relative prices (Buera and Kaboski, 2012). The work shows that manufacturing establishments in the U.S. have an average of 47 workers, compared to only 14 workers in services. Similarly, at the enterprise level, manufacturing firms employ 28 workers on average across OECD countries, while service firms only have 8 workers. Larger scale suggests higher fixed costs, as manufacturing requires substantial investment in machinery, infrastructure, and production facilities.

Profit in the Manufacturing Sector

$$\pi_M(k, l; R, w, p_M) = p_M z_M k^{\alpha_M} l^{\theta_M} - Rk - wl - (1 + r)\kappa_M \quad (6)$$

Profit in the Service Sector

$$\pi_S(k, l; R, w, p_S) = p_S z_S k^{\alpha_S} l^{\theta_S} - Rk - wl - (1 + r)\kappa_S \quad (7)$$

where:

p_j is the price of sector j output. In the original work, [Buera et al. \(2011\)](#) normalizes $p_M = 1$ for simplicity ⁶.

r is the interest rate.

κ_j represents the fixed operating cost in each sector.

2.1.3. Investment and Capital Choice

Entrepreneurs choose capital and labor to maximize their profits. In the absence of financial constraints, i.e., perfect capital markets where firms can borrow freely at the market interest rate R (which could be equal to $1 + r$ under certain settings). Thus, the optimal factor inputs for sector j solve ⁷:

$$\left(k_j^*(z_j), l_j^*(z_j) \right) = \arg \max_{k,l} \left\{ p_j z_j k^{\alpha_j} l^{\theta_j} - Rk - wl \right\} \quad (8)$$

Firms in the manufacturing and service sectors face different investment decisions. Hence, we provide the investment choice separately for each sector:

Manufacturing Sector

$$\max_{k,l} \left\{ p_M z_M k^{\alpha_M} l^{\theta_M} - Rk - wl - (1 + r)\kappa_M \right\} \quad (9)$$

Service Sector

$$\max_{k,l} \left\{ p_S z_S k^{\alpha_S} l^{\theta_S} - Rk - wl - (1 + r)\kappa_S \right\} \quad (10)$$

Nonconvexity and Scale Effects

In [Buera et al. \(2011\)](#)'s, fixed costs introduce nonconvexity in the production function. This means that firms must operate at a minimum scale to cover their fixed costs. The key condition for firm viability is:

$$z_j k_j^{\alpha_j} l_j^{\theta_j} \geq (1 + r)\kappa_j \quad (11)$$

where

⁶As mentioned in [2.1](#), the manufacturing sector acts as the numéraire, meaning that the manufacturing sector's price level is set to 1 to serve as the benchmark against which other prices are measured. Manufacturing is chosen because it tends to be more tradable internationally, meaning its prices are more stable and comparable across countries. Hence, setting $p_M = 1$ expresses the price of services relative to manufacturing and isolates and analyzes the impact of financial frictions on sectoral differences.

⁷If the model included borrowing constraints or required an upfront investment κ_j (e.g., a fixed setup cost for firms), then the financing constraint might appear explicitly. In such a case, the firm might need to borrow κ_j and repay it at rate $1 + r$, leading to an extra term $(1 + r)p_j \kappa_j$ in the budget constraint. However, the given problem assumes free access to capital markets, meaning firms can borrow as much as needed at rate R without additional fixed costs.

z_j = firm productivity in sector j

k_j, l_j are capital and labor used in sector j

α_j, θ_j are output elasticities of capital and labor

r is interest rate

This equation means that only firms with sufficient capital and productivity can enter the sector. If an entrepreneur cannot generate enough revenue to cover the fixed cost κ_j , they are forced to exit or not enter at all.

2.1.4. Credit and Rental Markets

The credit and capital rental market is the final player in the economy. Individuals interact with financial intermediaries, which are facilitators of capital allocation. These intermediaries accept deposits, rent capital k to entrepreneurs at a rental rate R , and provide loans to cover the fixed costs $p_j \kappa_j$ required for business operations.

Zero-Profit Condition in Financial Intermediation

The analysis assumes that all financial transactions, including borrowing and capital rental, occur within a single period. This restriction ensures that individuals' financial wealth remains nonnegative at all times ($a \geq 0$) to prevent long-term borrowing beyond the period.

Since financial intermediaries operate in a competitive market, they earn zero economic profit. This implies that the rental rate of capital must be equal to the sum of the interest rate on deposits (r) and the depreciation rate (δ):

$$R = r + \delta \quad (12)$$

An interpretation of the equation is that intermediaries break even when renting out capital and financial intermediaries cover the opportunity cost of funds (the interest rate r) and compensate for the loss of value in capital due to depreciation δ .

Imperfect Contract Enforcement and Default Risk

A critical friction in this financial market is the imperfect enforceability of financial contracts, which creates borrowing constraints. Specifically, after production takes place, entrepreneurs have the option to renege on their contractual obligations. If they choose to default, they can retain a fraction $(1 - \phi)$ of both (1) the undepreciated capital stock (after accounting for depreciation δ). and (2) the revenue from production, net of labor payments.

Formally, an entrepreneur who defaults can keep:

$$(1 - \phi) [p_j z_j f(k, l) - wl + (1 - \delta)k] \quad (13)$$

where

ϕ measures the strength of legal enforcement in financial markets ($\phi \in [0, 1]$). A higher ϕ indicates a more robust legal system that enforces financial agreements effectively, reducing the entrepreneur's ability to retain capital and revenue upon default⁸.

⁸ $\phi = 0$ means there is no contract enforcement, meaning entrepreneurs can fully renege on loans without consequences. This leads to a complete breakdown of credit markets. When $\phi = 1$, contracts are perfectly enforced, preventing entrepreneurs from defaulting and allowing financial markets to function efficiently.

$p_j z_j f(k, l)$ is the production output in sector j .

wl is the total labor payment.

δ is the depreciation rate of capital.

Since financial intermediaries cannot fully recover loans when defaults occur, they impose borrowing and capital rental limits on entrepreneurs.

Consequences of Default

The only penalty for defaulting is the seizure (garnishment) of the entrepreneur's financial assets a that were deposited with the intermediary. This means that even after defaulting, firms reset their reputation in the financial market in the next period and regain access to financial markets in the following period without additional restrictions.

This assumption implies a relatively weak enforcement environment, where the only deterrent against default is the potential loss of liquid financial assets, rather than broader repercussions such as exclusion from future borrowing or legal penalties. Consequently, financial institutions must design lending and capital rental contracts to be incentive-compatible so that entrepreneurs voluntarily comply rather than default.

Credit Constraints and Capital Rental Limits

To formalize these constraints, Buera et al. (2011) characterizes the equilibrium conditions that ensure borrowing and capital rental agreements remain enforceable. The enforceability of borrowing and capital rental contracts depends on the ability of financial intermediaries to ensure that entrepreneurs honor their obligations. Given that financial markets are imperfect due to contract enforcement frictions, i.e., the risk of default, intermediaries impose borrowing and capital rental constraints on entrepreneurs.

The upper bound $\bar{k}^j(a, z_j; \phi)$ is a sector-specific function that restricts the entrepreneur's capital rental based on their individual financial wealth a and productivity z_j . This constraint ensures that entrepreneurs do not have an incentive to default on their obligations, even when they can potentially retain a fraction $(1 - \phi)$ of the undepreciated capital and post-production revenue net of labor payments.

Since lending and rental agreements must remain viable under these conditions, financial intermediaries impose limits on capital allocation that differ across sectors:

$$\bar{k}^j(a, z_j; \phi) \leq k_u^j(z_j) \quad (14)$$

This condition is set as an assumption so that entrepreneurs cannot rent more capital than what is optimal in an unconstrained scenario. This guarantees that borrowing constraints are always binding for some entrepreneurs, thereby influencing sectoral investment dynamics.

In the next part, Proposition 1 formalizes the enforceability conditions for both manufacturing and service sectors.

Proposition 1: Condition for Contract Enforcement

We quote the Proposition statement of Buera et al. (2011)'s work as follows:

PROPOSITION 1. *Capital rental k in sector j by an entrepreneur with wealth a and talent z_j is enforceable if and only if*

$$\begin{aligned} & \max_l \{p_j z_j f(k, l) - wl\} - Rk - (1 + r)p_j \kappa_j + (1 + r)a \\ & \geq (1 - \phi) \left[\max_l \{p_j z_j f(k, l) - wl\} + (1 - \delta)k \right]. \end{aligned} \quad (15)$$

The upper bound on capital rental that is consistent with entrepreneurs choosing to abide by their contracts can be represented by a function $\bar{k}^j(a, z_j; \phi)$, which is increasing in a, z_j, ϕ .

The left-hand side (LHS) of equation (15) represents the economic return to the entrepreneur when they honor the rental contract. The right-hand side (RHS) shows the payoff if the entrepreneur defaults and refuses to honor the contract.

Proposition 1 states that entrepreneurs honor contracts only when the LHS is greater than or equal to the RHS. This means that **fulfilling the contract must be at least as profitable as defaulting**. The stricter the enforcement (higher ϕ), the smaller the fraction of retained output in case of default, making default less attractive.

Since entrepreneurs will rent capital only if their contracts are enforceable, financial intermediaries impose an upper bound on capital rental $\bar{k}^j(a, z_j; \phi)$. This function increases in the case that:

- Wealthier entrepreneurs (higher a) face a higher punishment if they default (because they lose all their deposits), making them more creditworthy.
- Wealthier entrepreneurs (higher talent z_j) face a higher punishment if they default (because they lose all their deposits), making them more creditworthy.
- Stronger legal institutions ϕ discourage default, allowing higher rental limits ⁹.

2.1.5. The Case of Perfect-Credit Market

When $\phi = 1$, financial markets function efficiently (see footnote 8). The transition from an imperfect credit market ($\phi < 1$) to a perfect-credit benchmark ($\phi = 1$) reflects the shift from financial frictions to an economy where all entrepreneurs have unrestricted access to capital for production within a period:

$$\bar{k}^j(a, z_j, \phi) = k_j^u(z_j), \quad \forall a$$

However, there is still no between-period borrowing or consumption insurance, so that entrepreneurs cannot transfer wealth across time or insure against shocks.

This explicit case shows how economic outcomes (e.g., output, factor shares, establishment size distributions) would look in the absence of financial frictions. It serves as a baseline to compare with economies where credit constraints exist ($\phi < 1$).

Key Assumptions

1. Entrepreneurial Talent Follows a Pareto Distribution

Entrepreneurial ability z_j (where $j = S, M$ for service and manufacturing sectors) follows a mutually independent Pareto distribution:

$$\eta, (z_S, z_M) \sim \eta^2 (z_S z_M)^{-(\eta+1)} \quad \text{for } z_j \geq 1$$

The Pareto tail parameter η determines the degree of inequality in entrepreneurial talent. A lower η means more large firms.

⁹Although the contract enforcement parameter ϕ is uniform across sectors, enforceable rental contracts vary between manufacturing and services due to differences in (1) Technology (e.g., capital intensity of production), (2) Output prices (sector-specific market conditions).

2. **Capital Rental Markets are Perfect**, meaning entrepreneurs can freely rent capital for production within a period. Since there are no financial frictions, credit constraints do not distort firm decisions.

Buera et al. (2011) formalize the closed-form expressions of the net sectoral output function, factor shares, and the establishment size distribution in the perfect-credit benchmark in Proposition 2.

PROPOSITION 2. Assume that entrepreneurial talents for the two sectors follow mutually independent Pareto distributions¹⁰ with the same tail parameter η , $(z_S, z_M) \sim \eta^2 (z_S z_M)^{-(\eta+1)}$ ¹¹ for $z_j \geq 1, j = S, M$, and that active entrepreneurs are a small fraction of the population¹². Then the output of a sector, net of fixed costs, equals

$$Y_j(K_j, L_j; N) = A_j N^{\frac{(1/\eta)}{(\alpha+\theta+1/\eta)}} K_j^{\frac{\alpha}{(\alpha+\theta+1/\eta)}} L_j^{\frac{\theta}{(\alpha+\theta+1/\eta)}} \quad (16a)$$

$$A_j = \frac{w + p_j \kappa_j (\alpha + \theta + \eta^{-1})}{w + p_j \kappa_j} \left[\frac{\eta(1 - \alpha - \theta)}{\eta(1 - \alpha - \theta) - 1} \right]^{\frac{1}{(1+\eta(\alpha+\theta))}} \left[\frac{w + p_j \kappa_j}{p_j(1 - \alpha - \theta)} \right]^{\frac{(1-\eta(1-\alpha-\theta))}{(1+\eta(\alpha+\theta))}} \quad (16b)$$

Proposition 2 states that under the perfect credit market, assuming that entrepreneurial talent $\mathbf{z} \sim \text{Pareto}(\eta)$, i.e., entrepreneurs are a small fraction of the population, thus most people are workers, and firm productivities are Pareto-distributed, then the sector-level production function that, while bearing a superficial resemblance to the standard neoclassical Cobb-Douglas form, fundamentally differs in its microeconomic underpinnings.

The output function (16a), although the functional form appears Cobb-Douglas, the elasticities of output with respect to capital and labor are not simply α and θ as in the standard model. Instead, they are scaled by the factor $\alpha + \theta + 1/\eta$, where η characterizes the thickness of the tail of the entrepreneurial ability distribution (modeled as Pareto). Consequently, the elasticities of output with respect to capital, labor, and population are

$$\frac{(1/\eta)}{(\alpha + \theta + 1/\eta)} \quad \text{for } N; \quad \frac{\alpha}{(\alpha + \theta + 1/\eta)} \quad \text{for } K; \quad \frac{\theta}{(\alpha + \theta + 1/\eta)} \quad \text{for } L$$

These expressions reflect the fact that in Buera et al. (2011)'s framework, entrepreneurs are heterogeneous and self-select into production based on their productivity and the presence of fixed costs. Because entrepreneurs earn rents from their superior abilities, factor payments (to labor and capital)

¹⁰The model assumes that entrepreneurial talents follow a Pareto distribution because this type of distribution is well-suited for describing heavy-tailed phenomena, where a small number of individuals (or firms) have disproportionately high productivity or wealth. According to Benhabib and Bisin (2018), empirics show that firm sizes and entrepreneurial productivity follow a power-law (Pareto-like) distribution. Deviations from mutual independence across sectors can lead to unrealistic implications, such as extreme differences in establishment sizes between manufacturing and services. The Pareto assumption enables the model to capture this high degree of inequality in firm sizes while maintaining analytical tractability. Since the distribution of firm sizes exhibits a thick right tail, this formulation helps explain why some firms grow significantly larger than others within a sector.

¹¹The assumption that entrepreneurial talents in the two sectors (service and manufacturing) follow mutually independent Pareto distributions with the same tail parameter η means that the productivity levels of entrepreneurs in each sector are randomly distributed but follow a heavy-tailed pattern, where a small number of entrepreneurs have very high productivity.

¹²This implies that most individuals in the economy do not run businesses but instead supply labor or capital. This assumption helps simplify the model by allowing sectoral output to be expressed in a tractable form.

do not exhaust total output¹³. As a result, the sum of factor income shares is strictly less than one, unlike in the standard neoclassical model where the sum of the elasticities equals one under constant returns to scale and no profits^{14 15}. This means the actual payments to capital are lower than the capital elasticity in the production function. Thus, the presence of fixed costs and entrepreneurial heterogeneity distorts the standard relationship between capital elasticity and capital income shares, making them dependent on sector-specific conditions. Moreover, the role of the population size N is modified. In contrast to the standard model—where output scales linearly with N when productivity per worker is constant—here output scales with $N^{\frac{(1/\eta)}{(\alpha+\theta+1/\eta)}}$. This non-linear scaling captures a selection effect: as the population increases, the economy draws more highly talented entrepreneurs from the upper tail of the distribution, raising aggregate productivity and output beyond simple input accumulation.

Proposition 2 also provides a formulation of the sector-specific productivity term A_j , which is endogenously determined by the interaction between wages w , fixed costs $p_j\kappa_j$, and the distribution of entrepreneurial talent on sectoral productivity.

The payments to capital in this setting are determined by the share of income allocated to capital in each sector. Unlike in a standard neoclassical model, where factor shares directly correspond to their elasticities in the production function, here the presence of entrepreneurial rents alters the distribution of income. The share of income paid to capital in sector j is denoted as

$$s_{K,j} = \frac{RK_j}{Y_j(K_j, L_j; N)} = \frac{\alpha(w + p_j\kappa_j)}{w + p_j\kappa_j(\alpha + \theta + \eta^{-1})} \quad (17)$$

Building on our previous result regarding sectoral output, we now turn to the distribution of establishment sizes within each sector. Given that entrepreneurial talents follow a Pareto distribution, we expect the resulting firm size distribution to exhibit a thick right tail, meaning a small number of firms account for a disproportionate share of employment and output.

Buera et al. (2011) formalizes these findings in Proposition 3:

¹³In the standard neoclassical growth model, output is entirely distributed to the factors of production—capital and labor—according to their marginal products. Mathematically, aggregate output Y satisfies $Y = wL + RK$. Under constant returns to scale and competitive markets, there are no pure profits or rents: each factor is paid exactly its marginal product, and there are no residual earnings left over. In contrast, in the Buera et al. framework, due to the presence of heterogeneous entrepreneurial talent, fixed costs of operation (κ_j), and endogenous selection into entrepreneurship, output is not entirely exhausted by payments to labor and capital. Highly talented entrepreneurs are able to generate sufficient output to not only cover factor payments and fixed costs, but also to earn positive rents. Formally, the capital and labor shares of output at the sectoral level are given by - Capital share = $\frac{RK_j}{Y_j}$, Labor share = $\frac{wL_j}{Y_j}$

However, due to the presence of entrepreneurial rents, these shares satisfy: $\frac{RK_j}{Y_j} < \alpha$; $\frac{wL_j}{Y_j} < \theta$. Hence, the sum of capital and labor share under Buera et al. (2011) is smaller than 1. Thus, the remaining share of output $1 - (\alpha + \theta) > 0$ goes to the entrepreneur as rents.

¹⁴In Buera et al. (2011)'s, $\alpha + \theta + 1/\eta$ is close to one so that the expressions for factor shares in the two sectors are approximately equal. Moreover, in a standard Cobb-Douglas production function, constant returns to scale require that the sum of the output elasticities equals one. Since entrepreneurs earn rents, the model does not impose strict constant returns to scale, but the similarity in factor shares suggests that deviations are small.

Another technical condition is that $\alpha + \theta + 1/\eta < 1$ to make sure that the model remains internally consistent, avoiding cases where increasing capital and labor input leads to disproportionate increases in output.

¹⁵This heterogeneity arises because entrepreneurs capture economic rents due to differences in productivity. In a perfectly competitive setting, factor payments exhaust total output, ensuring that factor shares sum to one. However, in this model, entrepreneurial talent follows a Pareto distribution, meaning some entrepreneurs are significantly more productive than others. These high-productivity entrepreneurs generate surplus profits beyond the marginal products of capital and labor, which they retain as rents. As a result, the capital and labor shares of output are influenced by the presence of fixed costs and entrepreneurial profits, leading to deviations from the standard neoclassical predictions.

PROPOSITION 3. Assume that entrepreneurial talents for the two sectors follow mutually independent Pareto distributions with the same tail parameter η , $(z_S, z_M) \sim \eta^2 (z_S z_M)^{-(\eta+1)}$ for $z_j \geq 1, j = S, M$, and that active entrepreneurs are a small fraction of the population. Then the establishment size distribution in each sector follows the power law

$$\Pr[\tilde{l}_j > l] = \left(\frac{l(\hat{z}_j)}{l} \right)^{\eta(1-\alpha-\theta)}, \quad l \geq l(\hat{z}_j) \quad (18)$$

where $l(\hat{z}_j)$ is the employment in the marginal establishment of sector j . Furthermore, the establishment size distribution in the aggregate economy is given by a mixture of Pareto distributions:

$$\Pr[\tilde{l} > l] = n_S \left(\frac{l(\hat{z}_S)}{l} \right)^{\eta(1-\alpha-\theta)} + n_M \left(\frac{l(\hat{z}_M)}{\max\{l, l(\hat{z}_M)\}} \right)^{\eta(1-\alpha-\theta)}, \quad l \geq l(\hat{z}_M) \quad (19)$$

where n_S and n_M are, respectively, the fraction of service and manufacturing establishments in the economy, with $n_S + n_M = 1$. Also, the ratio of the average establishment sizes of the two sectors is

$$\frac{\bar{l}_j}{\bar{l}_{j'}} = \frac{p_j \kappa_j + w}{p_{j'} \kappa_{j'} + w} \quad (20)$$

where

$\tilde{l}$ is firm size (employment) in sector j

$\hat{z}_j$ is the talent cutoff (minimum talent needed to operate a firm)

$l(\hat{z}_j)$ is the employment size of the marginal (smallest) active firm

$\bar{l}(\hat{z}_j)$ is a normalization constant related to that smallest firm size

Establishment Size Distribution in Each Sector

Under the assumptions of Proposition 3, each sector's establishment size distribution follows a power law because entrepreneurial talents z_j are drawn from a Pareto distribution with a common tail parameter η . In sector j , the probability that a firm's employment $\tilde{l}_j$ exceeds some level l is presented in equation (18), where $l(\hat{z}_j)$ is the size of the marginal entrepreneur's firm. This power-law form arises because: (1) the Pareto distribution of talents implies that extremely talented individuals are rare but possible, and (2) the model's production function, with diminishing returns (due to $\alpha + \theta < 1$), maps talent into firm size through a convex relationship - meaning that small differences in high talent levels translate into large differences in firm size. As a result, the upper tail of firm sizes becomes heavy and follows a Pareto law, with the exponent determined by both the talent distribution (η) and the returns to scale ($1 - \alpha - \theta$).

This means that larger firms are less frequent, but their distribution follows a power law, characterized by a tail exponent $\eta \cdot (1 - \alpha - \theta)$. This thick right tail aligns with empirical observations, where a small number of large firms coexist with a large number of small firms.

Aggregate Establishment Size Distribution

At the economy-wide level, the overall distribution of firm sizes is a mixture of the two sector-specific Pareto distributions, weighted by the share of establishments in each sector (n_S and n_M), as shown in equation (19), where $\max\{l, l(\hat{z}_M)\}$ ensures that the distribution is properly defined only for firm sizes above the minimum establishment size in manufacturing sector.

This result suggests that while both sectors follow a Pareto distribution, their composition in the aggregate economy matters. The dominance of one sector can affect the overall establishment size distribution.

Ratio of Average Establishment Sizes Across Sectors

Proposition 3 also compares sectoral firm sizes by relating them to sector-specific fixed costs and wages (20). High fixed costs act as a barrier to entry, restricting operation to only the most talented entrepreneurs who possess sufficiently high productivity levels. Consequently, when κ_j is large, only highly productive firms enter, leading to fewer but larger firms. Conversely, lower fixed costs allow a broader range of entrepreneurs to operate, resulting in more numerous but smaller firms.

This formulation provides a transparent mapping from observable data (such as firm size and wages) to fundamental structural parameters of the model. Because firm sizes and wages are typically available in data, this equation allows for indirect inference of otherwise unobservable sector-specific fixed costs. It thus connects microeconomic firm dynamics (entry, size distribution) with macroeconomic outcomes (sectoral output, productivity).

Overall, under the perfect credit benchmark, i.e., an economy with unconstrained within-period borrowing (without between-period borrowing or consumption insurance), firm-level decisions on capital and labor allocation are driven purely by entrepreneurial talent and sectoral technology, rather than financial frictions.

The net output in each sector follows a Cobb-Douglas production function, with sector-specific productivity and input elasticities determined by technological parameters and entrepreneurial talent distributions.

The Pareto distribution of entrepreneurial talent leads to a thick-tailed establishment size distribution, a well-documented feature in empirical data. This suggests that large firms emerge naturally due to differences in entrepreneurial ability rather than financing constraints.

Unlike standard neoclassical models, the elasticities of capital and labor in production are not equal to factor shares, as entrepreneurs earn economic rents due to their heterogeneous productivity levels. This distinction helps explain observed deviations in measured factor income shares from theoretical predictions.

The model establishes a direct link between sector-specific fixed costs and average firm size, providing a structural method for identifying key parameters using real-world data, i.e., an empirical bridge between firm-level heterogeneity and aggregate sectoral performance.

2.2. Individual's Optimization Problem

In section 2.1.2, we first assume households are born as entrepreneurs in one of two sectors: manufacturing (**M**) or services (**S**). In this section, we generalize the individual's problem to have them explicitly choose between being an employee or an entrepreneur in either sector.

State Variables and Occupational Choice

At the start of a period, each individual's state is defined by wealth a and talent $\mathbf{z}$. Given these state variables, the individual must decide whether to: (1) work as an employee (earning wages w , save money, and consume goods) or (2) become an entrepreneur in sector **S** (use wealth and talent to produce and sell service-sector goods) or sector **M** (use wealth and talent to produce and sell manufacturing-sector goods), paying a fixed cost κ_j (where $j = S, M$).

The value function $v(a, \mathbf{z})$ represents the maximum possible lifetime utility the individual can achieve given their current state, by choosing the best occupation:

$$v(a, \mathbf{z}) = \max \left\{ v^W(a, \mathbf{z}), v^S(a, \mathbf{z}), v^M(a, \mathbf{z}) \right\} \quad (21)$$

where

$v^W(a, \mathbf{z})$ is the value of working as an employee.

$v^S(a, \mathbf{z})$ and $v^M(a, \mathbf{z})$ are the values of being an entrepreneur in sectors S and M , respectively.

Since the entrepreneurial decision depends on the entire vector of talents $\mathbf{z}$, individuals can switch sectors later if it becomes more profitable. However, for simplicity of the analysis and model replication, we assume that households cannot switch away from the sector they are born into.

Employee's Problem

If an individual chooses to be a worker, their objective is to maximize utility (1) from consumption and future savings. In which, the decisions they need to consider are

1. A consumption bundle $\mathbf{c} = (c_S, c_M)$ that consists of goods from both sectors.
2. Next period's assets a' to maximize his continuation value, subject to the period budget constraint.

$$v^W(a, \mathbf{z}) = \max_{\mathbf{c}, a' \geq 0} u(\mathbf{c}) + \beta \left\{ \gamma v(a', \mathbf{z}) + (1 - \gamma) \mathbb{E}_{\mathbf{z}'} \left[v(a', \mathbf{z}') \right] \right\} \quad (22)$$

s.t. $\mathbf{p} \cdot \mathbf{c} + a' \leq w + (1 + r)a,$

where w is the labor income, $(1 + r)a$ is the return on assets, $\mathbf{p}$ is the vector of goods prices. γ is the probability that talent remains unchanged; with probability $1 - \gamma$, it follows a stochastic process $\mathbf{z}' \sim \mu(\mathbf{z}')$.

This means that a worker can accumulate wealth through savings, and in the next period, they will reconsider their occupational choice based on their updated wealth and talent.

Entrepreneur's Problem

The individual decides to become an entrepreneur if $v^j(a, \mathbf{z}) > v^W(a, \mathbf{z})$. Thus, they choose between the two sectors j (where $j = S, M$) by comparing expected profits and financial constraints. Individuals as entrepreneurs need to optimize their consumption $\mathbf{c}$, next period asset a' , and the production factor allocation of capital k and labor l to employ. This decision-making process is constrained by borrowing limits and the requirement that their business must generate enough profits to honor financial contracts.

1. Entrepreneur's Choice and Value Function

At the beginning of each period, an individual has wealth a (financial assets) and talent vector $\mathbf{z} = (z_S, z_M)$, where z_j determines productivity in sector j .

If they decide to be an entrepreneur, they choose whether to enter sector S (services) or sector M (manufacturing). The lifetime utility of an entrepreneur in sector j is represented by the value function:

$$v^j(a, \mathbf{z}) = \max_{\mathbf{c}, a', k, l \geq 0} u(\mathbf{c}) + \beta \left\{ \gamma v(a', \mathbf{z}) + (1 - \gamma) \mathbb{E}_{\mathbf{z}'} \left[v(a', \mathbf{z}') \right] \right\} \quad (23)$$

2. Constraints on the Entrepreneur's Problem

- (a) **Budget Constraint** From section 2.1.2, we have the production functions (3) and (4) of each sector. From those, we derived the profit functions (6) and (7). Since productivity $\mathbf{z}$ and output prices $\mathbf{p}$ differ across sectors, an individual will prefer the sector where expected net profits are higher.

$$\mathbf{p} \cdot \mathbf{c} + a' \leq p_j z_j f(k, l) - Rk - wl - (1 + r)p_j \kappa_j + (1 + r)a \quad (24)$$

where $p_j z_j f(k, l)$ is the Revenue from production, Rk is the capital rental costs (interest paid on capital rented), $(1 + r)p_j \kappa_j$ is fixed entry cost for entrepreneurship in sector j and $(1 + r)a$ is return on financial wealth. This condition states that total expenditures (consumption, savings, inputs) must not exceed total income (profits + financial returns).

- (b) **Capital Rental Limit**

$$k \leq \bar{k}^j(a, z_j; \phi) \quad (25)$$

Entrepreneurs cannot rent unlimited capital due to imperfect contract enforceability. The maximum capital they can rent depends on their own wealth a (acts as collateral), sector-specific talent z_j (determines profitability), and enforceability parameter ϕ .

We aggregate the factors (18), (19) and (20) and summarize the entrepreneur's problem as follows:

$$\begin{aligned} v^j(a, \mathbf{z}) = & \max_{\mathbf{c}, a', k, l \geq 0} u(\mathbf{c}) + \beta \left\{ \gamma v(a', \mathbf{z}) + (1 - \gamma) \mathbb{E}_{\mathbf{z}'} \left[v(a', \mathbf{z}') \right] \right\} \\ \text{s.t. } & \mathbf{p} \cdot \mathbf{c} + a' \leq p_j z_j f(k, l) - Rk - wl - (1 + r)p_j \kappa_j + (1 + r)a \\ & k \leq \bar{k}^j(a, z_j; \phi) \end{aligned} \quad (26)$$

3. The Market Equilibrium

A stationary competitive equilibrium is a state where the economy reaches a steady-state distribution of wealth and entrepreneurial abilities, such that all markets clear, financial contracts hold, and the decision-making behavior of individuals remains consistent over time.

Individual problem dynamics

The stationary competitive equilibrium consists of an invariant (i.e., *steady-state structure of inequality in both wealth and business potential*) distribution of wealth and entrepreneurial ideas $G(a, \mathbf{z})$, with the marginal distribution of $\mathbf{z}$ is derived from $G(a, \mathbf{z})$ by integrating out a (*to show how entrepreneurial ideas are distributed independent of wealth*).

Each individual in the economy makes decisions about consumption in each sector $c_S(a, \mathbf{z})$, $c_M(a, \mathbf{z})$, next period asset accumulation $a'(a, \mathbf{z})$, occupational choice $o(a, \mathbf{z})$ (whether to be a worker or entrepreneur), labor and capital input choices $l(a, \mathbf{z})$, $k(a, \mathbf{z})$ (if they are entrepreneurs in either sector).

These decisions depend on prices $(w, R, r, \mathbf{p})$ (5) and the borrowing limit $k_j(a, z_j; \phi)$ (14). The policy functions of consumption in each sector $c_S(a, \mathbf{z})$, $c_M(a, \mathbf{z})$, next period asset accumulation $a'(a, \mathbf{z})$, occupational choice $o(a, \mathbf{z})$, and factor allocation $l(a, \mathbf{z})$, $k(a, \mathbf{z})$ then are used to solve the optimization problems of (21), (22) and (23) (section 2.2).

Financial Markets and Capital Rental

Financial intermediaries, as key players in the economy, allocate capital efficiently so that there is no arbitrage. They provide capital rental but must break even, leading to the condition $R = r + \delta$ (12)

so that financial institutions cover depreciation costs while still earning a return equal to the risk-free rate.

Entrepreneurs require capital to operate, but not all of them have sufficient wealth to self-finance their businesses. Financial intermediaries step in to provide capital rental, allowing productive entrepreneurs to borrow based on their expected returns. The borrowing limits $\bar{k}^j(a, z_j; \theta)$ reflect financial frictions and credit constraints. However, these limits cannot exceed the capital demand of the most productive firms, given by $k_u^j(z_j)$, ensuring that resources are allocated efficiently among businesses: $\bar{k}^j(a, z_j; \theta) \leq k_u^j(z_j)$ (14).

The presence of borrowing constraints influences wealth accumulation and the composition of active entrepreneurs. Entrepreneurs with higher talent z_j but lower wealth may face tighter borrowing constraints, limiting their ability to expand their businesses. This, in turn, affects the overall efficiency of capital allocation, as some highly productive firms may remain undercapitalized due to credit frictions.

Market Clearing Conditions

Market clearing ensures that supply equals demand in all factor and goods markets. The key markets that must satisfy the conditions are:

1. Capital Rental Market:

$$\frac{K}{N} \equiv \int k(a, \mathbf{z}) G(da, dz) = \int a G(da, dz). \quad (27)$$

The left-hand side represents the total capital demand by entrepreneurs. The right-hand side represents the total wealth held by agents in the economy. Since financial intermediaries cannot create capital, the available capital supply must be fully allocated to entrepreneurs who rent it for production. If there are financial frictions (borrowing constraints), some entrepreneurs may not be able to rent the capital they need, leading to misallocation.

2. Labor Market:

$$\int l(a, \mathbf{z}) G(da, dz) = \int_{\{o(a, \mathbf{z})=W\}} G(da, dz). \quad (28)$$

The left-hand side represents the total labor hired by firms. The right-hand side represents the total number of workers who choose to remain in wage employment rather than becoming entrepreneurs. If wages are too high, fewer entrepreneurs will demand labor, leading to an excess labor supply. Conversely, if wages are too low, more individuals will not prefer wage employment, making it harder for firms to hire.

3. Services Market:

$$\int c_S(a, \mathbf{z}) G(da, dz) = \int_{\{o(a, \mathbf{z})=S\}} \left[z_S k(a, \mathbf{z})^\alpha l(a, \mathbf{z})^\theta - \kappa_S \right] G(da, dz). \quad (29)$$

4. Manufacturing Market:

$$\int c_M(a, \mathbf{z}) G(da, dz) + \delta K_N = \int_{\{o(a, \mathbf{z})=M\}} \left[z_M k(a, \mathbf{z})^\alpha l(a, \mathbf{z})^\theta - \kappa_M \right] G(da, dz). \quad (30)$$

Equations (29) and (30) represent total demand for services (for manufactured goods, which includes both consumption c_M and replacement investment due to capital depreciation δK_N). The right-hand side represents total output produced in the services (manufacturing) sector net of fixed costs. The condition ensures that all service and manufactured goods produced are either consumed (or used for capital replacement in the case of manufactured goods).

Stationarity Condition of the Wealth Distribution

The equilibrium requires that the joint distribution of wealth and entrepreneurial talent remains constant over time:

$$G(a, \mathbf{z}) = \gamma \int_{\{(a', \mathbf{z}') | \mathbf{z}' \leq \mathbf{z}, a'(\mathbf{z}') \leq a\}} G(da', dz') + (1 - \gamma) \mu(\mathbf{z}) \int_{\{(a', \mathbf{z}') | a'(\mathbf{z}') \leq a\}} G(da', dz'). \quad (31)$$

The first term represents entrepreneurs who continue operating. The second term represents new entrants who replace exiting entrepreneurs. The economy reaches a stationary equilibrium when the distribution of wealth and entrepreneurial talent no longer changes over time.

4. Implementation - Decentralized Model with Occupational Choice

The original model by Buera et al. (2011) features a two-sector economy in which households choose between wage employment and entrepreneurship across distinct sectors. However, for this analysis, we consider a simplified version of the model that operates in partial equilibrium and collapses the economy into a single sector. This simplification enables us to focus more precisely on the core dynamics of occupational choice and household decision-making without the additional layer of complexity introduced by inter-sectoral differences.

In our version, factor prices such as the interest rate and wage rate are taken as exogenous, and the model does not impose market-clearing conditions. By isolating the key mechanisms in a single-sector setting, we aim to better understand how asset holdings and productivity influence occupational choices and consumption-savings behavior.

Model setup The model is solved as a decentralized problem where infinitely-lived households must choose between being workers or entrepreneurs. The population is normalized to 1, $N_t = 1$, and some fraction of these are workers, $L_t < N_t$.

Household's problem The household's problem when they are workers is

$$\begin{aligned} & \max_{\{c_t\}_{t=0}^{\infty}, \{a_{t+1}\}_{t=0}^{\infty}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \frac{c_t^{1-\sigma}}{1-\sigma}, \\ & \text{s.t.} \quad c_t + a_{t+1} = (1+r)a_t + w_t, \\ & \quad c_t > 0, \\ & \quad a_t \text{ is given.} \end{aligned} \quad (32)$$

where w_t is the wage rate that households take as given and the constraints hold for $t = 1, 2, \dots$

Entrepreneurs's problem The household's problem when they are entrepreneurs is

$$\begin{aligned} & \max_{\{c_t\}_{t=0}^{\infty}, \{a_{t+1}\}_{t=0}^{\infty}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \frac{c_t^{1-\sigma}}{1-\sigma}, \\ & \text{s.t.} \quad c_t + a_{t+1} = (1+r)a_t + \pi_t, \\ & \quad c_t > 0, \\ & \quad a_t \text{ is given.} \end{aligned} \quad (33)$$

where π_t is the profit of the firm and the constraints hold for $t = 1, 2, \dots$. Firms are in perfectly competitive markets and so take prices as given. **The firm's problem is static** and given by

$$\begin{aligned} \max_{K_t L_t} e^{\varepsilon_t} K_t^\alpha L_t^{1-\alpha} - w_t L_t - R_t K_t - (1 + r_t) \kappa, \\ \text{s.t. } K_t \leq \bar{K}(a_t, \varepsilon_t, \phi), \end{aligned} \quad (34)$$

where $\varepsilon_t \sim \mathcal{N}(0, \sigma_\varepsilon^2)$ and the rental limit, $\bar{K}(a_t, \varepsilon_t, \phi)$, satisfies

$$\max_{L_t} e^{\varepsilon_t} K_t^\alpha L_t^{1-\alpha} - w_t L_t - R_t K_t - (1 + r_t) \kappa + (1 + r) a_t \geq (1 - \phi) \left[\max_{L_t} e^{\varepsilon_t} K_t^\alpha L_t^{1-\alpha} - w L_t + (1 - \delta) K_t \right] \quad (35)$$

The intuition in the constraint above is that firms will not default on their rental contracts for capital if the profit from doing so exceeds the profit if their assets, excluding undepreciated capital, are garnished. The model is solved in partial equilibrium, so r and w are set exogenously and do not adjust to equate supply and demand. Similarly, L_t is set exogenously, which firms take as given, and K_t is obtained from the first order conditions of the firm, given fixed w_t, r_t, δ , and K_t .

At each period, households compare the value of being a worker versus an entrepreneur and choose the occupation that maximizes expected utility. The occupational choice function thus reflects how asset levels and productivity influence the switching between employment types.

5. Results

5.1. Baseline policy functions of occupational choice and consumption

We examine the baseline policy functions that govern household occupational choices and consumption decisions. In each period, households evaluate the expected lifetime utility associated with becoming a wage worker versus operating as a self-employed entrepreneur. This comparison depends on their current state—specifically, asset holdings and productivity levels—and determines their optimal occupational status. The resulting occupational choice function maps the regions in state space where each occupation is preferred.

Occupational choice policy function Figure 1 illustrates how households decide between self-employment and wage work based on their current asset holdings and productivity levels.



Figure 1: Optimal Occupational Choice (0 as Self-Employed, 1 as Worker)

In general, households with higher productivity and greater asset holdings are more likely to choose self-employment (*yellow region* = 0). This is because of their greater capacity to bear risk and invest in entrepreneurial activity. Conversely, households with low assets or low productivity opt for wage work, valuing its income stability and lower exposure to shocks (*blue region* = 1). We see the importance of initial conditions and access to resources in shaping labor market outcomes. The model captures the intuitive result that wealth and talent jointly determine occupational sorting: self-employment becomes attractive only when households have sufficient buffers or capabilities to manage entrepreneurial risk.

Optimal consumption Simultaneously, conditional on occupational status, households make intertemporal consumption decisions, as shown in Figure 2.

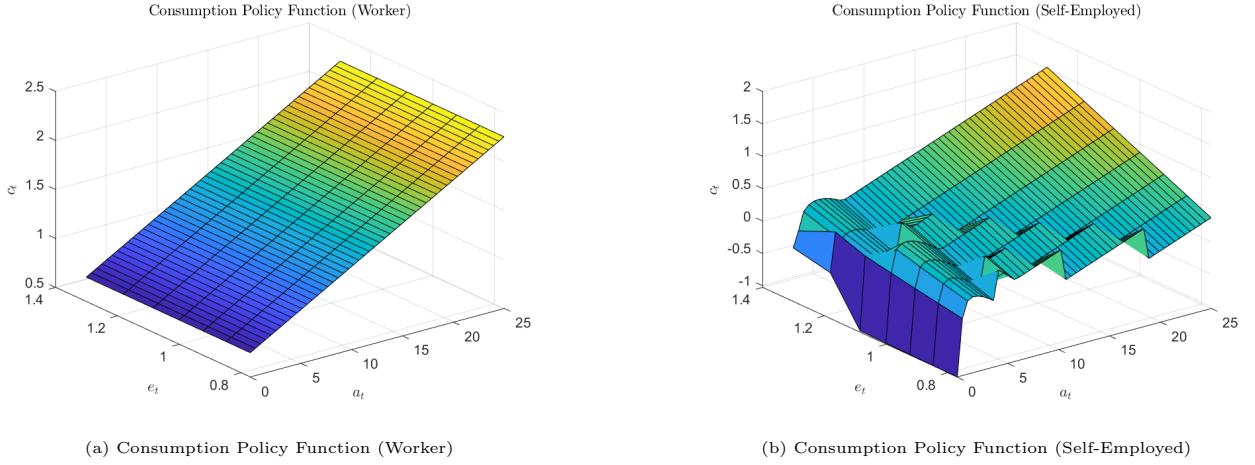


Figure 2: Consumption Policy Function of (2a) Worker, and (2b) Self-Employed Individuals

For workers (Figure 2a), the consumption surface is smooth, continuous, and monotonically increasing in both asset levels and productivity. This suggests that under wage employment, households engage in economically intuitive and stable consumption-savings behavior (see figure 29a for workers' optimal saving behavior)¹⁶. We usually see such regularity in consumption patterns in standard life-cycle consumption theory, where higher wealth and earning potential afford households greater consumption opportunities. For workers, the wage-earning environment is characterized by stable, predictable income streams and relatively low exposure to risk.

However, there are significant irregularities in the consumption policy function for the self-employed (Figure 2b). We see that the consumption surface includes abrupt discontinuities, jagged regions, and even negative consumption values in some parts of the state space, particularly at low levels of assets and productivity. Economically, these irregularities can be partly rationalized by the inherent volatility and nonlinear payoff structure of entrepreneurial activity. Unlike wage earners, whose income is relatively stable and predictable, self-employed agents face idiosyncratic risk, uncertain returns to effort, and high exposure to fixed costs or capital constraints. These conditions naturally introduce non-convexities in the value function, which can lead to kinks or discontinuities in the associated policy functions. For instance, small changes in asset holdings or productivity might cause a discontinuous jump in optimal consumption if they trigger shifts in credit access, profitability, or participation constraints.

¹⁶For optimal saving policy functions of workers and self-employed individuals, see Appendix A, figure 29

5.2. Change in technological parameters

Building on the baseline model, we now examine how key technological parameters shape household decisions and macroeconomic outcomes. Specifically, we consider how changes in the real wage, interest rate, and capital depreciation rate affect occupational choices, consumption, savings, and entrepreneurial behavior. These parameters serve as structural features of the economic environment, governing the returns to labor and capital and influencing the constraints and incentives faced by heterogeneous households.

5.2.1. Changes in real wages

First of all, we have chosen to assess the effects that different values of wages have on our model. Wage levels have an effect on both the employee's and the entrepreneur's value function. It directly affects the state of the employee, as wage levels affect savings levels, and therefore affects the choices that they make for occupational choices, consumption, and savings for the next period. Wage levels also affect the decisions of entrepreneurs, as these individuals have to make payments to labor, and their profit from operating is affected.

Therefore, we have decided to examine the economy across three different scenarios: the baseline case with $w = 0.75$, a higher wage of $w = 0.85$, and an even higher case with $w = 1.0$, from left to right. The reason that we chose not to go with a case for a lower wage level than the baseline case is that in the baseline case, the wages are low enough that many individuals with limited talents and assets can become entrepreneurs. Therefore, choosing higher wage levels to examine can offer more meaningful scenarios where the changes in the policy functions are more pronounced.

Value Policy Functions The worker value function graphs from figure 3 demonstrate a clear upward shift in lifetime expected utility as real wages increase.

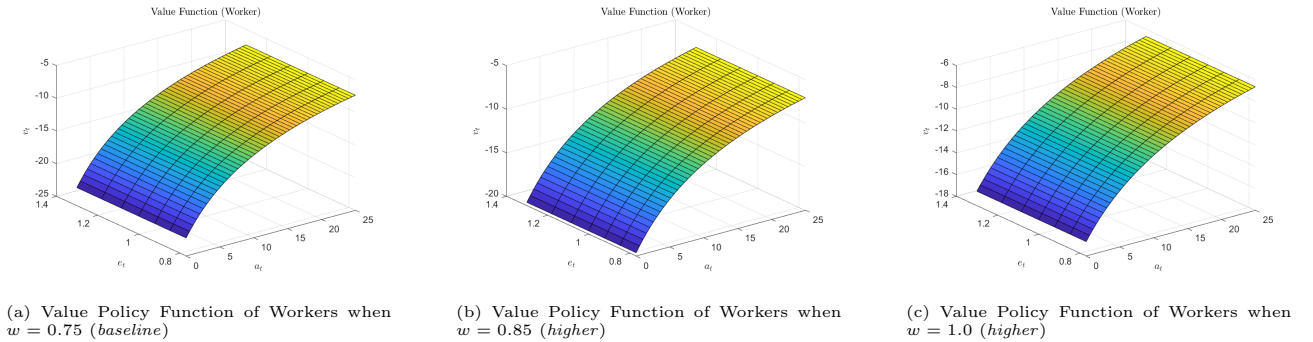


Figure 3: Value Policy Function of Workers for different wage rates

At the baseline wage level ($w = 0.75$), the value function of workers ranges from roughly -25 to -7 (Figure 3a), but as wages rise to 0.85 (Figure 3b) and then to 1.00 (Figure 3c), the value surface lifts, with the range beginning much higher, as high as -18 at the lower-asset end. Although the surface shape and gradient remain consistent—rising steadily with asset holdings and showing only modest sensitivity to productivity—the overall elevation in value indicates improved welfare for workers at all levels of productivity and wealth. The diminished responsiveness of the value function to productivity confirms that asset holdings, which enable greater consumption smoothing, are the primary determinant of worker welfare in this setting.

Occupational Choices The patterns of workers' value policy functions provide meaningful insight into occupational sorting in response to wage changes. Figure 4 illustrates occupational choices across

three scenarios: (a) the baseline case with $w = 0.75$, (b) $w = 0.85$, and (c) $w = 1.00$, respectively.



Figure 4: Occupational Choice (0=Self-Employed, 1=Worker) for different wage rates

As wages increase across Figure 4, the region for wage workers (*yellow areas in the plot*) expands, implying that higher wages make being a worker more desirable. This is because as wages increase, the value function for being an employee increases, while the value function for being an entrepreneur decreases due to higher payments to labor. This graphical representation of an individual's occupational choices aligns with our predictions for the model.

Consumption for Wage Workers and Self-Employed Figure 5 and 6 illustrate consumption policy functions for workers and entrepreneurs, respectively. The graphs show significant contrasts between workers and self-employed individuals in how they respond to wage increases.

For workers (Figure 5), consumption changes minimally across different wage rates ($w = 0.75$ (fig. 5a), 0.85 (fig. 5b), and 1.0 (fig. 5c)).

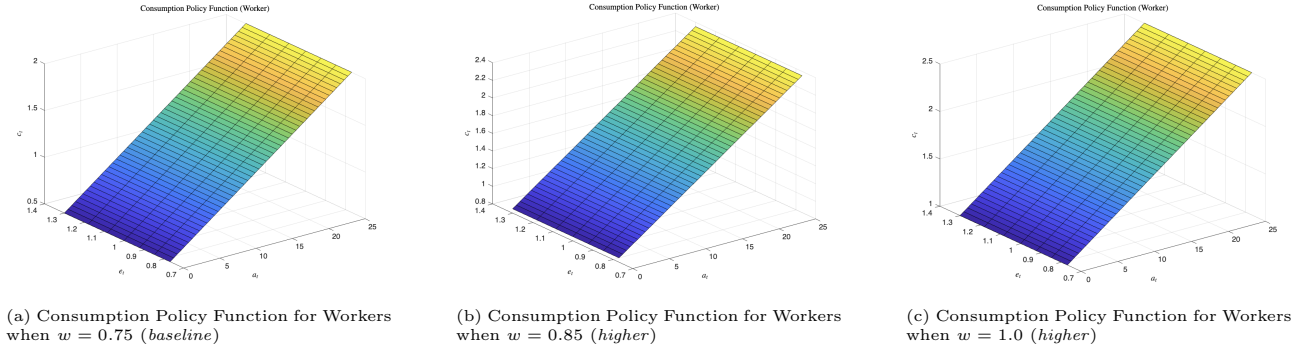


Figure 5: Consumption Policy Function for workers for different wage rates

For workers, the budget constraint is given by $c_t + a_{t+1} = (1 + r)a_t + w_t$ (equation (32)). This linear constraint implies that consumption increases with assets and wages. However, the observed changes in worker consumption are minimal—despite rising wages—because these wage adjustments are relatively small compared to total income, especially for asset-rich individuals whose interest income $(1 + r)a_t$ may dominate. Additionally, workers face less income risk and optimize according to the permanent income hypothesis, which smooths consumption over time. As a result, small increases in current wages have little effect on lifetime expected income, leading to nearly flat consumption responses across wage scenarios.

In contrast, self-employed households exhibit substantial consumption volatility in response to wage increases. Their consumption surfaces are irregular, with visible discontinuities and threshold effects linked to occupational choice and capital rental constraints.

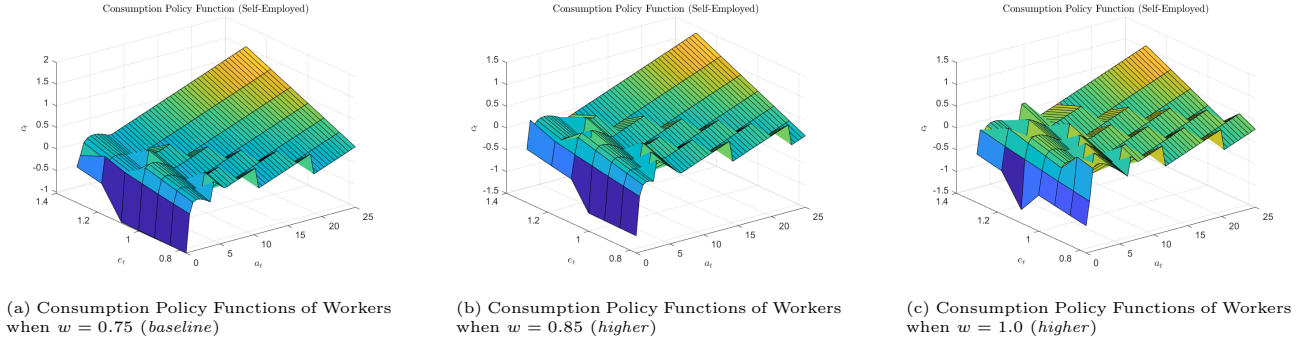


Figure 6: Consumption Policy Functions of Workers for different wage rates

Self-employed individuals face a more volatile budget constraint $c_t + a_{t+1} = (1 + r)a_t + \pi_t$ (equation (33)), where profit π_t is endogenously determined and highly sensitive to factor prices like wages. As wages rise, labor becomes more expensive, directly reducing π_t and thereby shrinking disposable income. The resulting non-linear and discontinuous consumption surfaces reflect this volatility, as entrepreneurs respond to reduced profits by cutting consumption rather than savings (see Appendix B.1, figure 30 and 31) —likely to maintain future investment potential.

Capital Rental Limit Capital rental limits do not change as wages increase because wages do not directly affect the entrepreneur’s collateral or the lender’s risk assessment (See Appendix B.1, figure 32). In the model, the capital rental limit is typically tied to the entrepreneur’s ability to repay - that is, the value of collateral (e.g., undepreciated capital) and the potential output or profit from using that capital. Since labor is a variable input and wages are a cost rather than an asset, an increase in the wage rate raises operating costs but does not reduce the collateral value or directly change the capital stock’s physical or depreciated value.

Profit Policy Function Examining the profit policy function across wage levels of 0.75, 0.85, and 1.0 in figure 32 reveals a consistent downward shift in profits as wages rise.

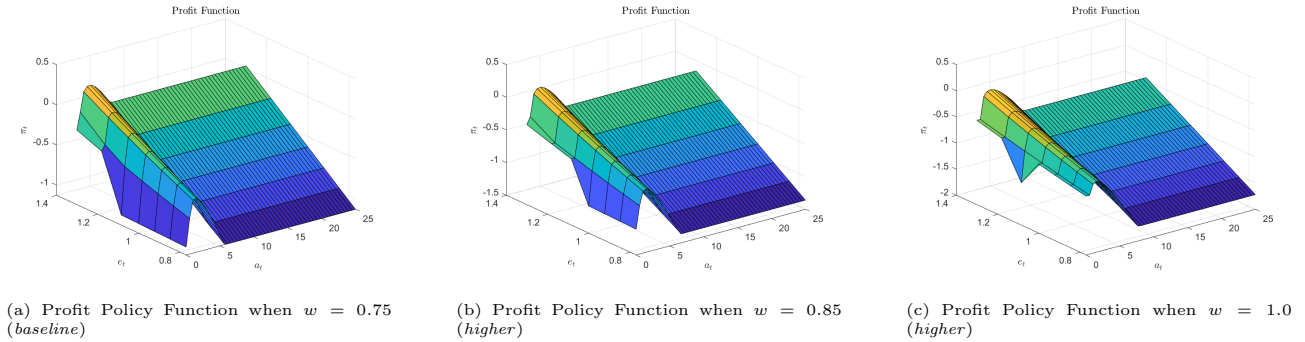


Figure 7: Profit Policy Function of Workers for different wage rates

While the general shape of the surface remains stable—profits increase with both productivity and assets—the overall levels drop notably. Maximum profits decline modestly, while minimum profits fall more sharply, suggesting increasing risk or loss potential for entrepreneurs (equation (34)). Moreover, the gradient of the profit surface with respect to productivity flattens at higher wages, indicating that gains from productivity improvements are less rewarding in high-wage environments. Discontinuities in the profit surface also become more pronounced, highlighting sharper transitions in entrepreneurial viability around key productivity-asset thresholds.

These shifts can be traced to higher labor costs directly reducing profits under the assumption of perfect

competition, where output prices are fixed and cannot adjust to absorb cost increases. Entrepreneurs partially respond by adjusting their labor demand, but because the marginal product of labor must equal the higher wage, labor usage contracts unless matched by high productivity. Additionally, wealthier entrepreneurs are better able to substitute capital for labor, softening the impact of rising wages. However, this substitution is insufficient to maintain profitability for most entrepreneurs. As a result, higher wages both lower entrepreneurial returns and raise the relative appeal of wage work, reinforcing the observed decline in self-employment as wages increase.

5.2.2. Changes in interest rate

Next, we have chosen to assess the effects that different values of interest rates have on our model. In the model construction, we see that setting interest rates affects asset accumulation for both the employee's and the entrepreneur's problem (equation (32), (33)). Besides that, the fixed cost (κ_j) also evolves at a rate of $(1 + r)$, which coincides with the interest rate r in our model (equation (11)). Therefore, adjusting the interest rate r will consequently have on the savings level in the economy, and therefore other decisions will follow. Moreover, this change in interest rate will make being an entrepreneur harder, as the fixed cost increases. Finally, varying interest rates also affect the rental rate for capital (R), as illustrated in equation (12). Higher R values, due to higher r , can lead to capital being more expensive to rent, thus affecting the decision to become entrepreneurs. Examining the occupational choices for the individuals will be crucial to see if increasing interest rates have a more substantial effect on asset accumulation or the barrier for entry into entrepreneurship.

There are 3 cases for examination: $r = 0.06$ (baseline case), $r = 0.04$ (lower case), and $r = 0.085$ (higher case).

Value Policy Functions of Workers and Self-Employed Figures 8 and 9 illustrate the value functions of being a worker and an entrepreneur in different cases of interest rates value r , respectively.

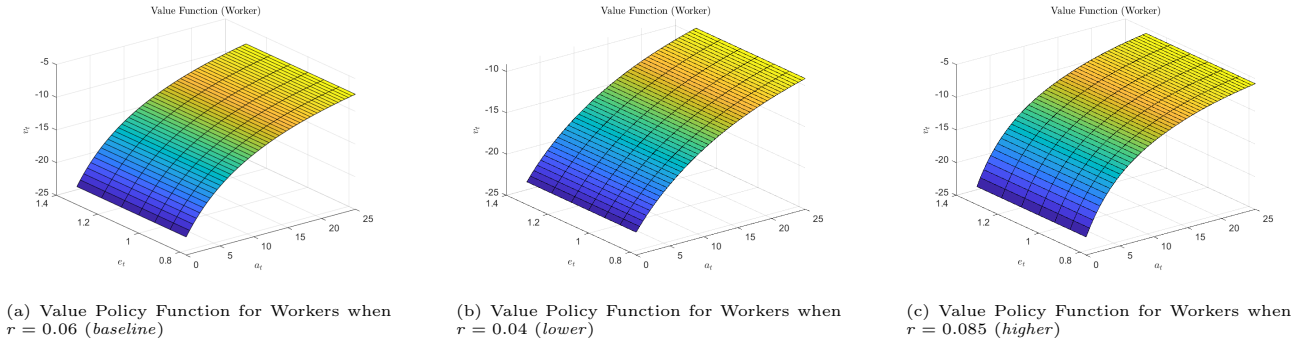


Figure 8: Value Policy Function for workers for different interest rates

Overall, we can notice a similar trend for the value function of workers, as observed in Figures 8a, 8b, 8c. However, we notice that for cases of higher asset levels for the three graphs, they differ in the perceived values. For example, in Figure 8b, for people with the highest asset levels, the perceived utility is approximately -10. However, for higher interest rates in Figures 8a and 8c, their values for individuals with the highest asset levels almost reached -5. We can see that for higher interest rates, workers in the economy have a higher value function, and this effect is visible for workers with higher asset levels.

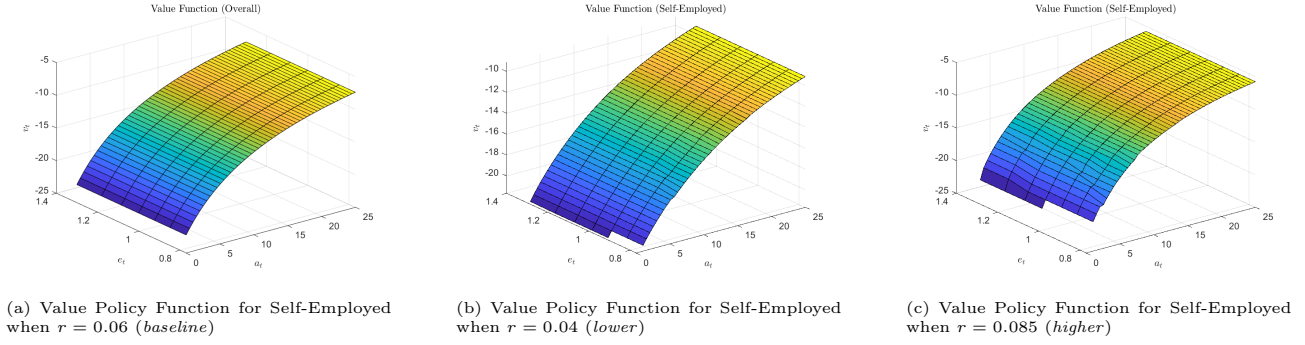


Figure 9: Value Policy Function for Self-Employed for different interest rates

For entrepreneurs in our economy, the value functions also show the same patterns as for workers. However, for Figures 9b and 9c, we notice that for individuals with the lowest assets and talents, their value function returns $-\infty$ values due to negative consumption.

Occupational Choice Figure 10 illustrates the occupational choices for individuals in our economy in different cases of interest rate value r , respectively.



Figure 10: Occupational Choice (0=Self-Employed, 1=Worker) for different r

Overall, from Figures 10a, 10b, and 10c, we can see the shift for individuals to become workers more in the economy. From the lower case in Figure 10b, we can see that people with the lowest levels of both assets and talents want to become workers. This suggests that because the fixed cost κ_j evolves at a smaller rate, the barrier for entering entrepreneurship is quite low. However, as r increased, it became harder to be entrepreneurs, hence the more choices for being wage workers in Figures 10a and 10c. Because the asset accumulation happens for both being a wage worker and an entrepreneur, the difference in occupational choices might be due to the involvement of the fixed cost κ_j and capital rental rate R , which are only incurred once self-employed. This trend for occupational choice aligns with the value function part discussed above.

Consumption Policy Functions for Workers and Self-Employed Figure 11 and 12 illustrate the consumption policy functions of being a worker and an entrepreneur in different cases of interest rate value r , respectively.

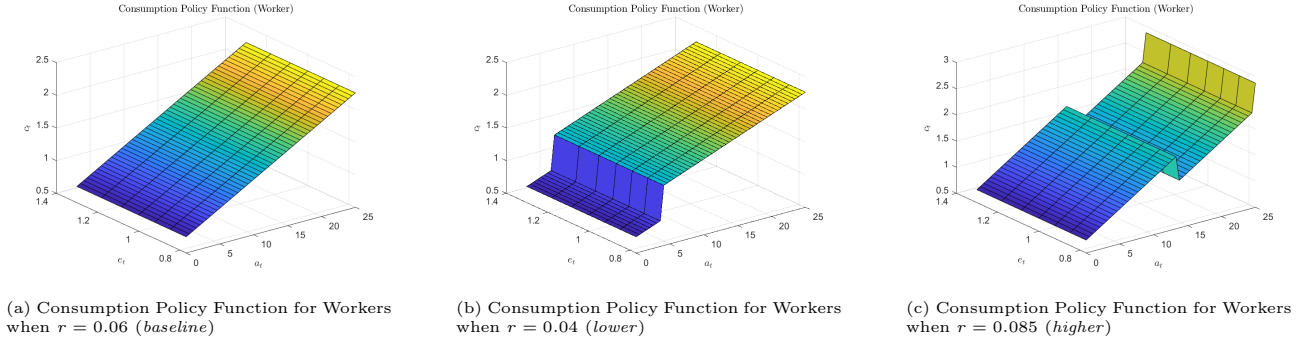


Figure 11: Consumption Policy Function for workers for different interest rates

Observing the consumption behaviors in Figures 11a, 11b, and 11c, we could notice some very different trends among these three graphs. When setting $r = 0.06$, the consumption increases steadily as we move up to individuals with more assets (see Figure 11a). However, when setting a lower $r = 0.04$, we can notice a steep jump in consumption level when individuals have $a \approx 5$. Then, as assets increase even more, consumption levels also rise gradually, peaking at $c_t \approx 2.3$ when assets are at their highest levels. Finally, when setting $r = 0.085$, there is a drop in consumption levels when $a_t \approx 14$. However, when asset levels are the highest, consumption for individuals in this economy reaches the peak of almost 3. This suggests that on aggregate, workers in the economies with higher interest rates consume the most, which aligns with the value functions in Figure 8.

Below are the policy functions for entrepreneurs in our economy, which explain why, as interest rates increase, more individuals choose to be workers:

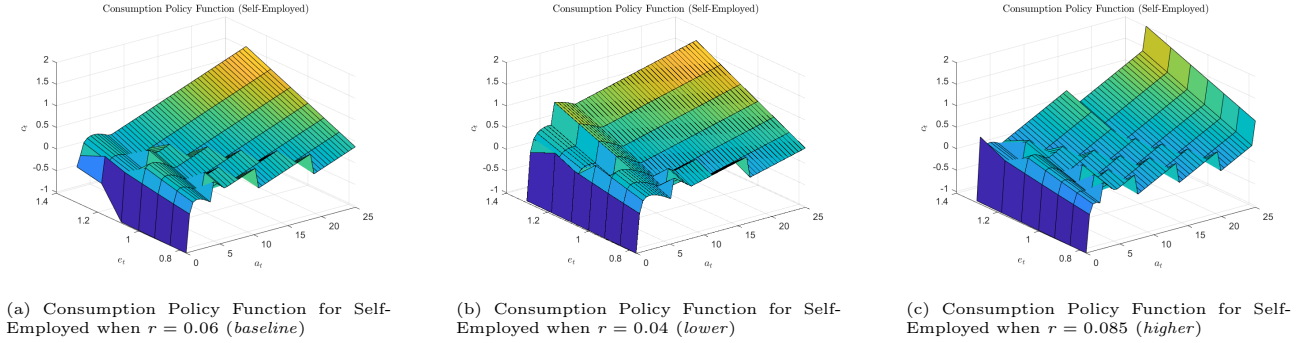


Figure 12: Consumption Policy Function for Self-Employed for different interest rates

Overall, for Figures 12a, 12b, and 12c, there are some noticeable trends. The consumption levels are mostly, on aggregate, lower than for workers, holding talents and wealth constant. Even when wealth and talents are at the highest levels, consumption for entrepreneurs does not reach 2. Besides that, when assets and talents are low, consumption for entrepreneurs is even below zero, which explains the $-\infty$ gap in the value functions in Figure 9.

We also notice that when asset levels are high, a higher interest rate helps asset accumulation to outweigh the effect of the rise of the fixed cost κ_j and the rental rate R . Hence, when assets are high, people in the economy with higher interest rates consume more than in economies with lower interest rates. However, when the asset levels are low, the effect of the entry barriers is relatively more substantial than asset accumulation. Thus, when assets are low, people in the economy with higher interest rates consume less than in economies with lower interest rates. This creates a situation where lower-interest-rate economies may actually lead to higher consumption for individuals with low wealth.

Savings Policy Functions for Workers and Self-Employed Figure 13 and 14 illustrate the savings policy functions of being a worker and an entrepreneur in different cases of interest rate value r , respectively.



Figure 13: Savings Policy Function for workers for different interest rates

Overall, the savings policy for workers for three values of r , illustrated through Figures 13a, 13b, and 13c, aligns with the consumption policy functions in Figure 11. Savings when $r = 0.06$ increases steadily with asset level, as the same trend is observed with consumption in Figure 11a. Savings when $r = 0.04$ has a drop when $a_t \approx 5$, which aligns with the steep jump in consumption at the same level of assets in Figure 13b. Finally, when $r = 0.085$, savings have a jump when $a_t \approx 14$, which also aligns with the steep drop in consumption at the same level of asset, as illustrated in Figure 13c. The saving-consumption trade-off for workers in our economy is clearly illustrated in the policy functions in Figures 11 and 13.

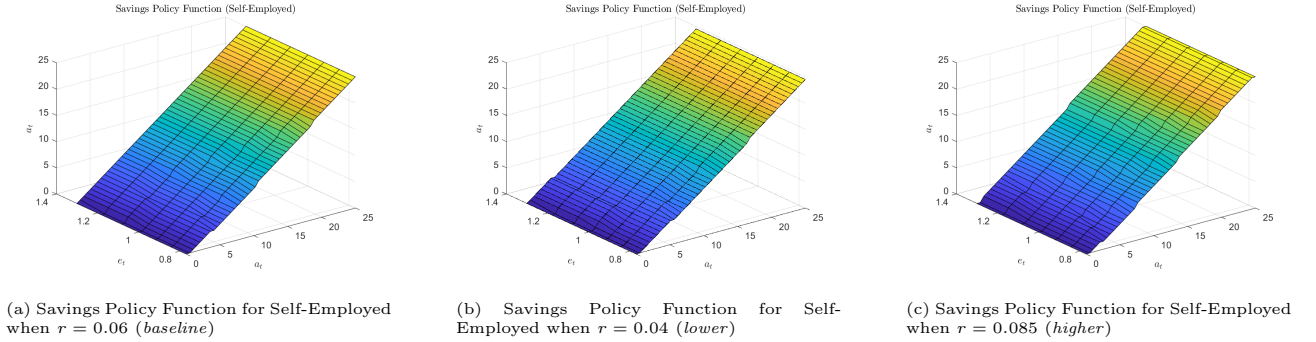


Figure 14: Savings Policy Function for Self-Employed for different interest rates

Overall, these savings policy functions also align with the consumption policy functions for entrepreneurs, in Figure 12. When compared with Figure 13, we can observe that the savings trends are quite similar. However, across individuals with different talents, their saving levels slightly fluctuate, but these effects are quite small.

To sum up, both the savings of workers and entrepreneurs, in Figures 13 and 14, illustrate the consumption and savings for the next period trade-off relationship in equations (22) and (24). Across economies with different interest rates, the consumption and savings levels also behave in the way that we predicted.

Capital Rental Limit Figure 15 illustrates the capital rental limit for being an entrepreneur in different cases of interest rate value r , respectively.

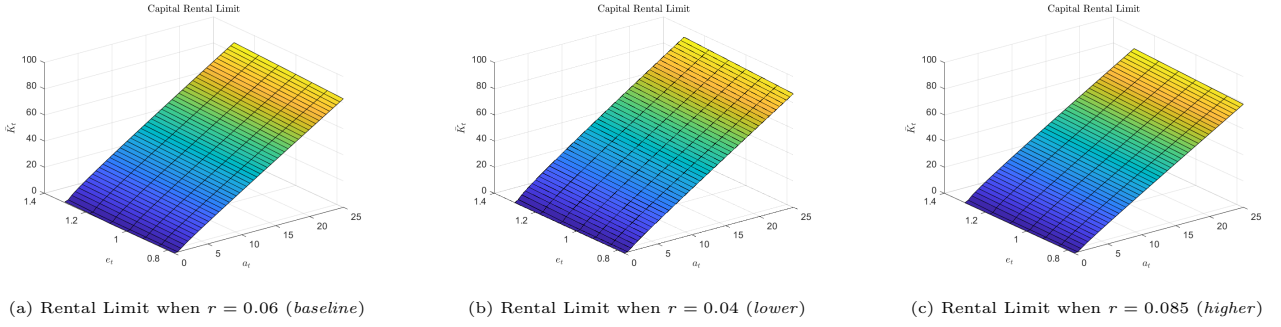


Figure 15: Rental Limit for different interest rates

We can notice a trend in Figure 15: As the interest rate r is increased, the capital rental limit is slightly decreased. This is because when the interest rate is low, capital is cheaper to rent (R is smaller), and the bound limit is increased, as illustrated through equations (8) and (14). Conversely, when capital is more expensive to rent, due to higher R from higher r , the capital rental limit is decreased. Therefore, the visual representation of the capital rental limit aligns with our theories.

Profit Policy Function Looking at the profit policy functions in Figure 16, we can notice that the profits for entrepreneurs decrease as the interest rate is increased.

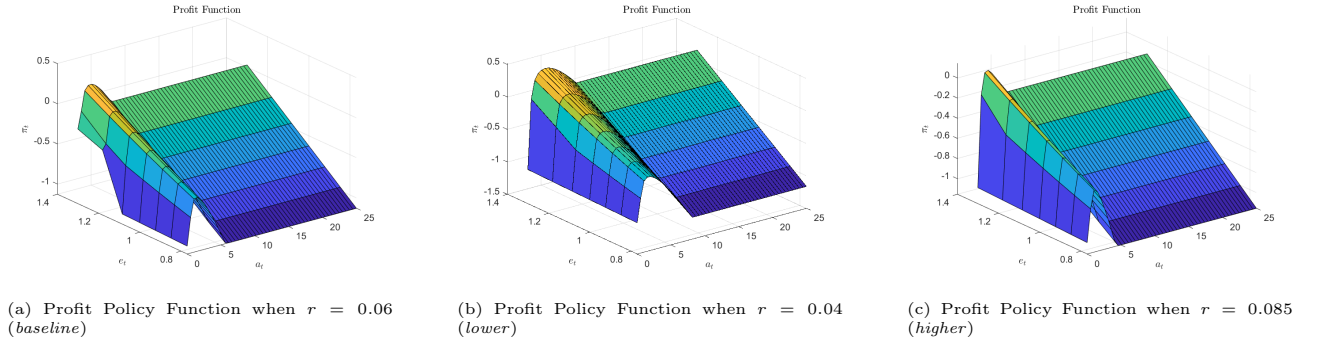


Figure 16: Profit Policy Function for different interest rates

This aligns with our theories, as a lower r results in a lower capital rental rate (R) and fixed costs for operation (κ_j). These decreased parameters R and κ_j make the profits for entrepreneurs increase (see equation (5)). This effect is illustrated as such: when considering the same level of assets and talents, most of the time, individuals in the economy with the lower interest rate would yield higher profits than the economy with the higher interest rate.

5.2.3. Changes in depreciation rate

As discussed in section 5.1, when asset holdings increase, households are able to pursue self-employment at lower productivity levels, suggesting that wealth serves as a buffer against the risks of entrepreneurship. This is evident across all three models, but the degree to which asset holdings influence the policy functions also varies depending on the depreciation rate.

To examine how capital depreciation affects household decision-making, we analyze how the policy functions respond to changes in the depreciation rate by varying depreciation δ at a lower depreciation rate $\delta = 0.01$ and a higher depreciation rate $\delta = 0.05$ compared to the baseline value $\delta = 0.03$.

Occupational choices Figure 17 illustrates the results across three scenarios: (a) the baseline case with $\delta = 0.03$, (b) a lower depreciation rate $\delta = 0.01$, and (c) a higher depreciation rate $\delta = 0.05$, from left to right.

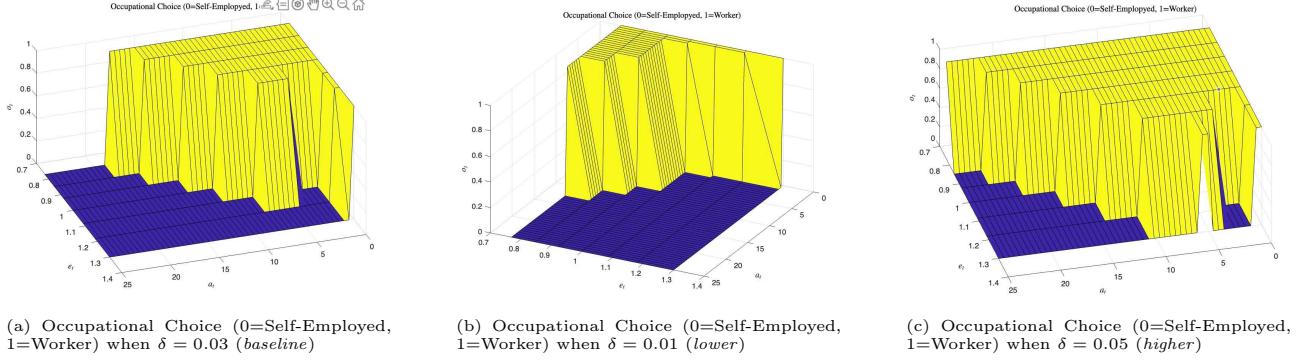


Figure 17: Occupational Choice (0=Self-Employed, 1=Worker) for different δ

As depreciation increases (Figure 17c), the region of wage employment (*yellow areas in the plot*) expands, implying that higher capital costs make entrepreneurship less viable. Specifically, a higher δ increases the asset threshold required for households to consider self-employment, as the net returns on entrepreneurial investments are diminished. In contrast, a lower δ reduces the asset threshold (Figure 17b), enabling households with lower assets to engage in self-employment due to lower capital costs and higher potential returns on investment.

Consumption Followed by, Figure 18 shows entrepreneur' optimal consumption c_t varies with productivity e_t and asset holdings a_t .

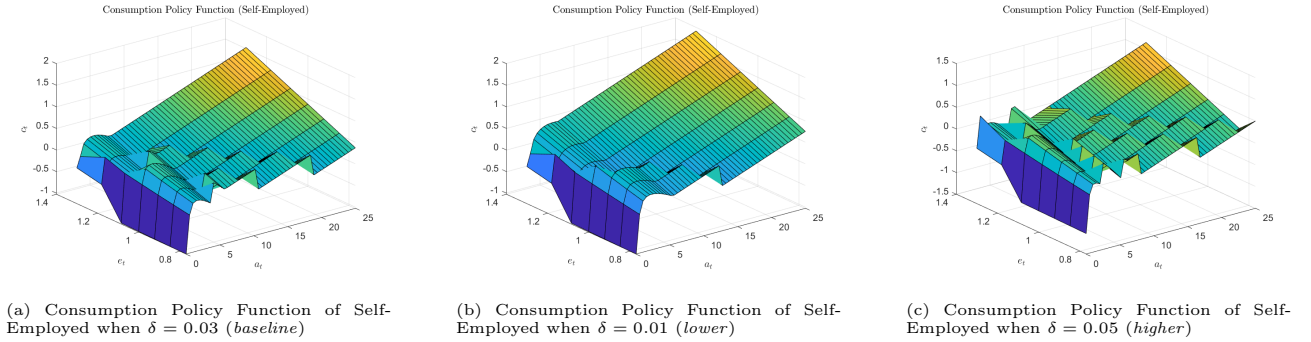


Figure 18: Consumption Policy Function of Self-Employed for different δ

Compared to the baseline scenario (Figure 18a), when the depreciation rate is low ($\delta = 0.01$), the consumption surface is smooth, and households exhibit higher overall consumption levels. Consumption rises steadily with increases in productivity and assets, indicating that lower capital replacement costs allow entrepreneurs to allocate more resources toward current consumption. At a high depreciation rate ($\delta = 0.05$), the consumption function becomes the most volatile, with clear drops at certain asset-productivity thresholds. Households reduce consumption more aggressively, likely in response to elevated uncertainty and reduced net returns from entrepreneurship.

These consumption patterns reflect the economic trade-offs faced by self-employed households as depreciation changes. Higher depreciation increases capital replacement costs, reducing disposable income and making entrepreneurial activity riskier. In response, households reduce consumption to save more, ensuring they can maintain their productive assets.

This contrasts sharply with wage workers, whose consumption remains largely unaffected by changes

in depreciation because their income does not rely on capital (See Appendix B.2, figure 33). Their consumption decisions are driven by wages and interest rates, which remain constant in this partial equilibrium model. The divergence in consumption responses underscores a key feature of the model: self-employed individuals bear capital-related risks, while wage workers face more stable, predictable consumption paths.

Saving policy function Similarly, only the savings of the entrepreneur respond to the intensity of the capital depreciation, as shown in figure 19. However, the magnitude is not significant and require further statistics to quantify the effect on the saving surfaces.

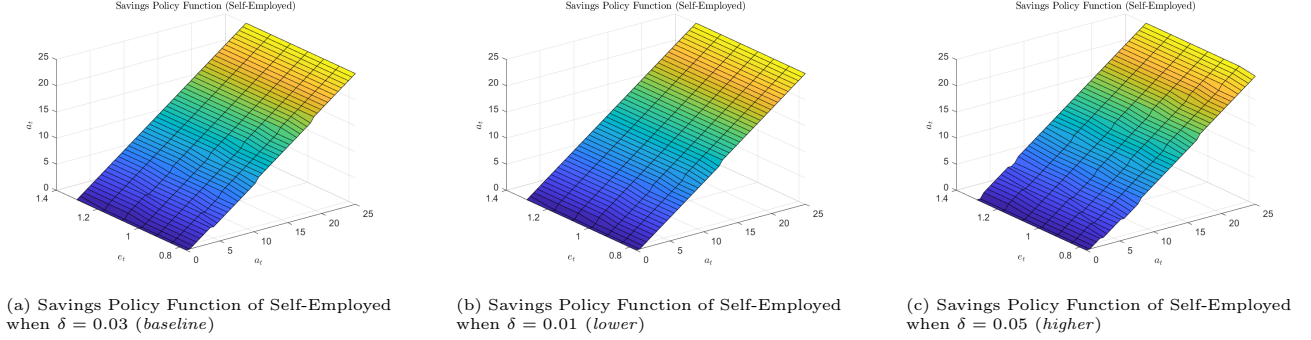


Figure 19: Savings Policy Function of Self-Employed for different δ

Rental limit $\bar{k}$ Examining the capital rental limit policy functions across different depreciation rates (Figure 20), while the overall shape of the surface remains similar, the maximum attainable capital rental limit $\bar{k}$ decreases as depreciation increases.

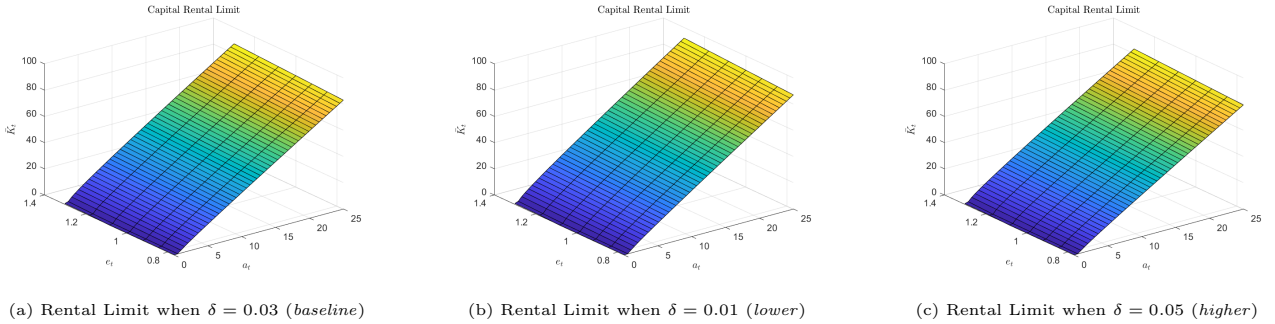


Figure 20: Rental Limit for different δ

In the low depreciation case ($\delta = 0.01$) in Figure 20b, entrepreneurs can access a higher maximum rental limit, reflecting greater borrowing capacity compared to the base case (Figure 20a). In the high depreciation scenario ($\delta = 0.05$), the ceiling of $\bar{k}$ is noticeably lower. Although the gradient of the surface—with $\bar{k}$ rising in response to higher asset holdings and productivity—remains consistent across cases, the upper bound is more restricted when depreciation is high.

The economic logic behind this pattern is expressed in equation (35), in which depreciation forms collateral constraints and investment returns. A higher depreciation rate directly reduces the present value of undepreciated capital $(1 - \delta)k$, weakening the collateral available to secure rental capital and increasing the likelihood of default. In response, lenders impose stricter borrowing limits, thereby lowering the rental limit $\bar{k}$. Furthermore, the reduced expected return on capital in high-depreciation settings discourages lending and amplifies the risk of overextension. As a result, the capital rental constraint becomes more binding, limiting the scale at which entrepreneurs can operate.

Profit Policy Function Finally, the profit policy function graphs in figure 21 reveal systematic changes in entrepreneurial profitability as the depreciation rate varies.

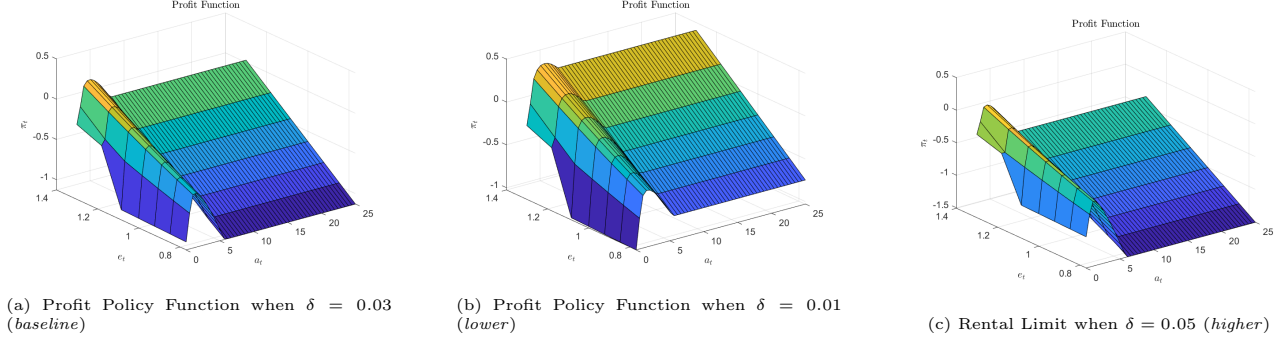


Figure 21: Profit Policy Function for different δ

At low depreciation ($\delta = 0.01$), profits span a wider and higher range (approximately -1.0 to 0.5), while in the baseline case ($\delta = 0.03$), this range contracts slightly (-1.0 to 0.3). Under high depreciation ($\delta = 0.05$), profits are notably lower across the board, ranging from -1.5 to just 0.1 . In all scenarios, profits increase with both productivity and asset holdings, but the rate of increase is steeper when depreciation is low. Discontinuities in the profit surface—where profitability drops sharply—often correspond with constraints on capital rentals, reinforcing the link between access to capital and entrepreneurial outcomes.

These patterns reflect the economic forces underlying the firm’s static profit maximization problem (Equation (34)). Higher depreciation directly raises the cost of using capital, as it is embedded in the effective rental rate. With output prices fixed (due to perfect competition), rising input costs compress profit margins. Moreover, higher depreciation tightens capital rental constraints, reducing the amount of capital entrepreneurs can access. This limitation disproportionately affects high-productivity households that would otherwise operate at a larger scale. Assets help relax these constraints, so the profit-asset relationship becomes more critical when depreciation is high. Additionally, fixed costs create a profitability threshold, and under higher depreciation, more state combinations fall below this threshold. Thus, depreciation has a dual effect: it reduces profitability through higher costs and indirectly suppresses scale through tighter capital access. This helps explain why entrepreneurship becomes less appealing under high depreciation, consistent with the occupational choice and consumption behavior observed in earlier analyses.

5.2.4. Conclusion on the effect of technological parameters

In summary, we have demonstrated the effects of key technological parameters—namely, the real wage w , interest rate r , and capital depreciation rate δ —on household behavior and economic outcomes within the model. Variations in these technological parameters reveal the sensitivity of household economic decisions to the structure of returns in labor and capital markets.

First, an increase in the *real wage* makes wage employment more attractive relative to self-employment, particularly for households with low productivity or limited asset holdings. As a result, the occupational threshold shifts, reducing the prevalence of entrepreneurship and concentrating it among high-wealth, high-productivity individuals.

Second, the *interest rate* shapes both intertemporal consumption choices and capital access. A higher interest rate incentivizes savings by increasing the return on deferred consumption, yet it simultaneously raises the cost of borrowing, tightening credit constraints for entrepreneurs. This dual effect leads to reduced consumption, constrained capital rental, and ultimately lower profitability and scale among self-employed households. In contrast, a lower interest rate environment eases credit conditions, allow-

ing households—especially those who are liquidity-constrained—to invest more in productive assets, smooth consumption, and increase future returns.

Third, the *capital depreciation rate* has a pronounced influence on entrepreneurial dynamics. A higher depreciation rate increases the effective cost of capital, reduces net returns from self-employment, and weakens the collateral value of existing capital stock. This restricts capital rental access, compresses profit margins, and elevates cautious saving behavior, especially among entrepreneurs. In contrast, lower depreciation improves capital efficiency, lowers investment costs, and enhances both profitability and operational scale, thereby enabling broader participation in self-employment, even among households with modest assets.

5.3. Change in Entry/Exit Criterion on Self-Employed

Fixed cost κ , as discussed in section 2.1.3, is the minimum condition scale of firm viability so that firms can enter the sector ($z_j k_j^{\alpha_j} l_j^{\theta_j} \geq (1+r)\kappa_j$). When the fixed cost of operating a business κ increases (equation (11)), the entry threshold for entrepreneurship becomes more stringent. This shift in the entry/exit adjusts the boundary between worker and entrepreneur in the state space, thus affecting individuals' optimal behaviors. Similar as section 5.2, We observe the magnitude of the responses of entrepreneurs by varying κ when $\kappa = 0.2$ and $\kappa = 0.85$ compared to the baseline value $\kappa = 0.5$.

Value Function of Self-Employed Figure 22 illustrates how the value function for self-employed individuals responds to changes in fixed business costs κ , thus explaining for the shifts observed in occupational choice patterns later on.

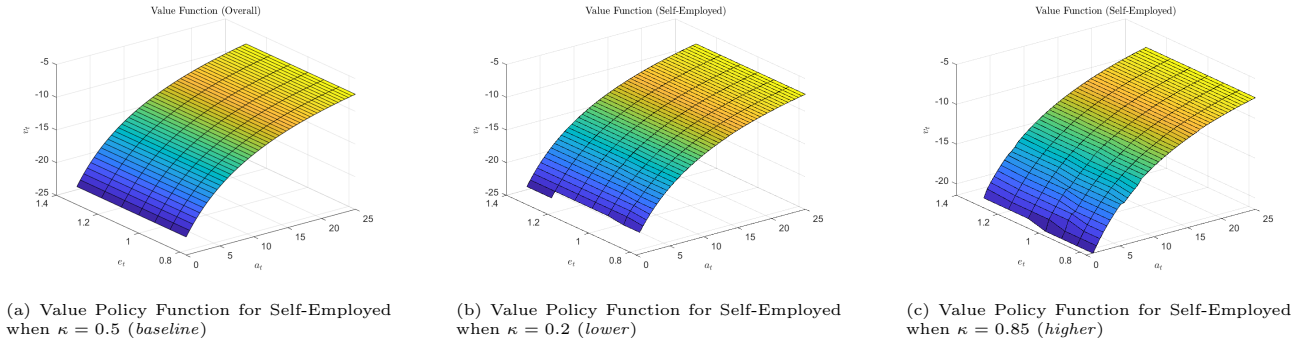


Figure 22: Value Policy Function for workers for different fixed costs

When fixed costs are reduced (fig. 22b, $\kappa = 0.2$), the value surface shifts upward, especially in areas with moderate productivity and asset levels compared to the baseline case (fig. 22a, $\kappa = 0.5$). This suggests that more individuals find self-employment worthwhile under lower entry barriers, consistent with the expansion of self-employed regions in the occupational choice graphs.

Conversely, increasing fixed costs (fig. 22c, $\kappa = 0.85$) leads to a downward shift in the value function, reducing the utility associated with entrepreneurship across much of the state space. This aligns with a contraction in the range of individuals who choose self-employment, as more agents find wage work comparatively more valuable.

Occupational Choice The direct relationship between fixed costs and the self-employment value function helps explain occupational sorting in the model, as shown in Figure 23.



Figure 23: Occupational Choice (0=Self-Employed, 1=Worker) for different κ

When fixed costs are lowered (fig. 23b, $\kappa = 0.2$), the yellow self-employment region expands noticeably compared to the baseline scenario (fig. 23a, $\kappa = 0.5$). This indicates that a wider range of individuals, including those with lower productivity or ability, now find self-employment profitable, as the entry barrier has been reduced. The threshold that separates self-employment from wage work shifts downward, making entrepreneurship a more accessible option.

By contrast, increasing fixed costs (fig. 23c, $\kappa = 0.85$) leads to a contraction of the yellow region. The higher costs raise the threshold required for entrepreneurship to be viable, forcing many individuals who would have chosen self-employment under lower costs to instead opt for wage work. This reallocation underscores how fixed costs influence the minimum ability or productivity required to sustain a business. Overall, this figure highlights how fixed costs in determining occupational sorting: lower costs democratize access to entrepreneurship, while higher costs restrict it to a more select group.

Profit Policy Function Fixed costs directly and inversely affect firms' optimal profit. And the effects are straightforward, as shown in Figure 24.

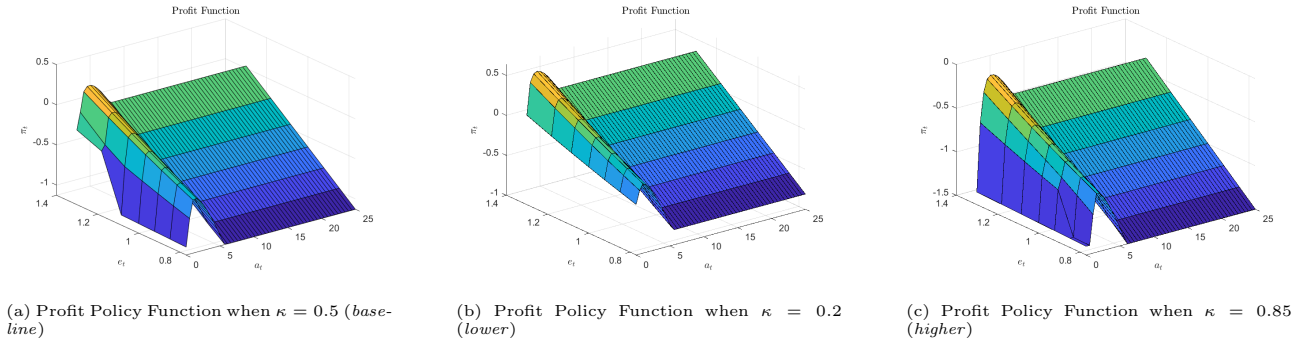


Figure 24: Profit Policy Function for different fixed costs

In the baseline scenario (24a, $\kappa = 0.5$), the profit surface is moderately elevated. A clear profit threshold separates viable self-employment from non-viable cases. When fixed costs are reduced (24b, $\kappa = 0.2$), the entire profit surface shifts upward. More combinations of the state variables (e.g., entrepreneurial ability κ_1 and productivity q) now yield positive profits, expanding the green/yellow region. This expansion reflects increased feasibility of entrepreneurship for individuals with more modest abilities, corresponding to the enlarged self-employment region in the corresponding occupational choice plots.

In contrast, when fixed costs increase (24c, $\kappa = 0.85$), the profit surface shifts downward, and green/yellow regions shrink. The threshold for viable entrepreneurship moves inward, meaning only individuals with high enough ability or productivity can cover the now larger fixed cost and still generate profits.

From the firm's static profit function, fixed costs reduce profits via the $(1 + r)\kappa$ term, raising the bar that entrepreneurial revenues must clear. Since capital use is constrained by $K \leq \bar{K}(a, \varepsilon, \phi)$, firms cannot scale arbitrarily to absorb higher fixed costs. As a result, marginal entrepreneurs—those near the threshold—are increasingly pushed into wage employment as κ rises.

Consumption From the earlier shifts observed in the profit function, consumption patterns for self-employed individuals change in response to variations in fixed business operating costs κ , as shown in Figure 25.

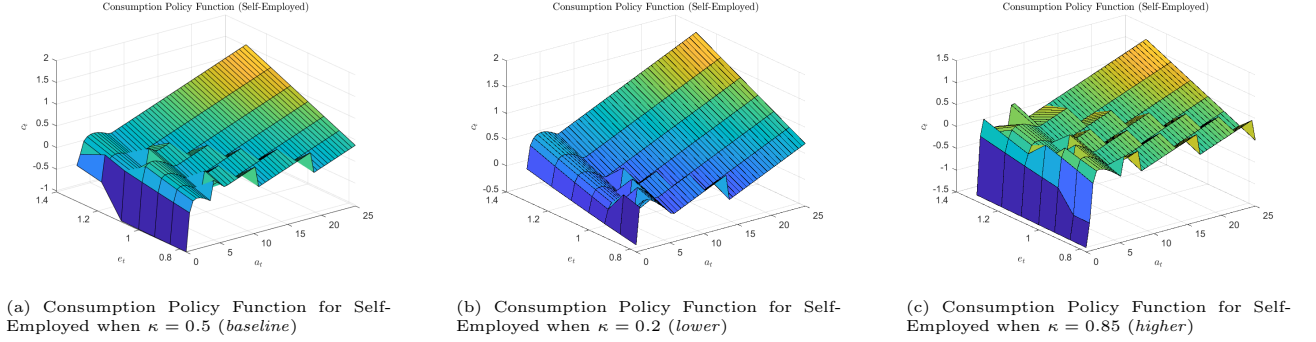


Figure 25: Consumption Policy Function for workers for different fixed costs

In the baseline case (fig. 25a, $\kappa = 0.5$), the consumption surface exhibits moderate variation across the state space, with higher consumption corresponding to regions of greater entrepreneurial ability (κ_1) and productivity (q). Discontinuities in the surface indicate thresholds where entrepreneurship becomes just viable, leading to sharp changes in consumption behavior. When fixed costs are reduced (fig. 25b, $\kappa = 0.2$), the entire consumption surface shifts upward. Entrepreneurs retain more of their profits due to lower costs, allowing for increased consumption across most of the state space. This aligns with the expanded profitability region seen earlier and the broader self-employment choice set. Conversely, when fixed costs rise (fig. 25c, $\kappa = 0.85$), the consumption surface drops noticeably. Even highly capable entrepreneurs face reduced disposable income, and pronounced discontinuities emerge, particularly near occupational thresholds. In some areas, consumption barely increases with ability, suggesting entrepreneurs in those regions may be better off switching to wage work.

This response in consumption is rooted in the entrepreneur's intertemporal budget constraint, $c_t + a_{t+1} = (1 + r)a_t + \pi_t$, which directly links current consumption to entrepreneurial profits. As fixed costs increase, profits π_t decline (as shown in the static firm problem where profits are reduced by $(1 + r_t)\kappa$), tightening the entrepreneur's constraint. Entrepreneurs are thus forced to either consume less or reduce future asset holdings. Given their long-term optimization goals, most adjust both margins.

5.4. Change in Capital Rental Limit

The rental limit $\bar{K}(a_t, \varepsilon_t, \phi)$ determines how much capital entrepreneurs can rent, directly affecting their production scale, profits, and occupational choices. In the model, capital rental limit $\bar{K}$ is endogenously determined by ϕ , changes in this institutional parameter directly affect entrepreneurs' ability to rent capital. Thus, we offer a mechanism of changing the rental limit $\bar{K}$ by varying $\phi \in [0, 1]$, i.e., changing the strength of contract enforcement, in two values $\phi = 0.2$ and $\phi = 0.9$ in comparison to the baseline $\phi = 0.75$.

Rental Limit Figure 26 captures changes in the strength of contract enforcement ϕ affect entrepreneurs' capital rental limits across the state space.

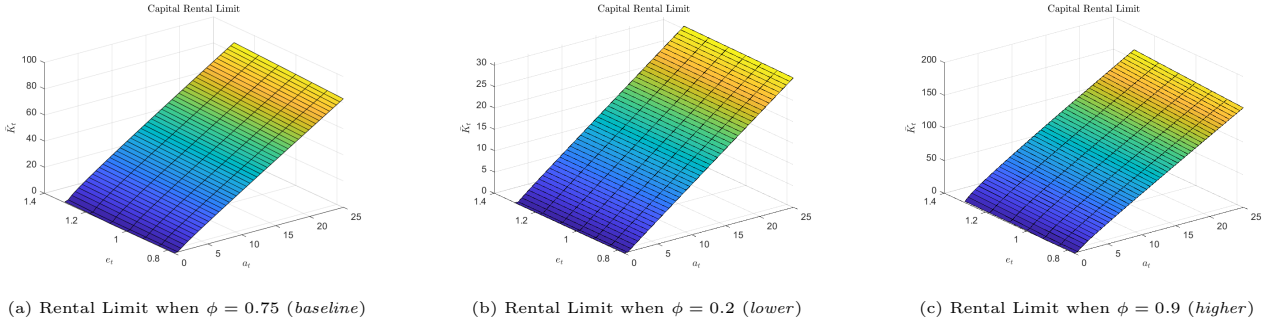


Figure 26: Rental Limit for different ϕ

In the baseline case (fig. 26a, $\phi = 0.75$), the rental limit ranges from 0 to 100 units, with higher ability individuals able to rent more capital. When contract enforcement weakens (fig. 26b, $\phi = 0.2$), rental limits drop sharply to a 0–30 range, compressing firm scale even for high-ability entrepreneurs. Conversely, stronger enforcement (fig. 26c, $\phi = 0.9$) lifts limits up to nearly 200 units, allowing top entrepreneurs to scale up significantly.

Profit Policy Function Correspondingly, profit function changes with ϕ , as shown in Figure 27.

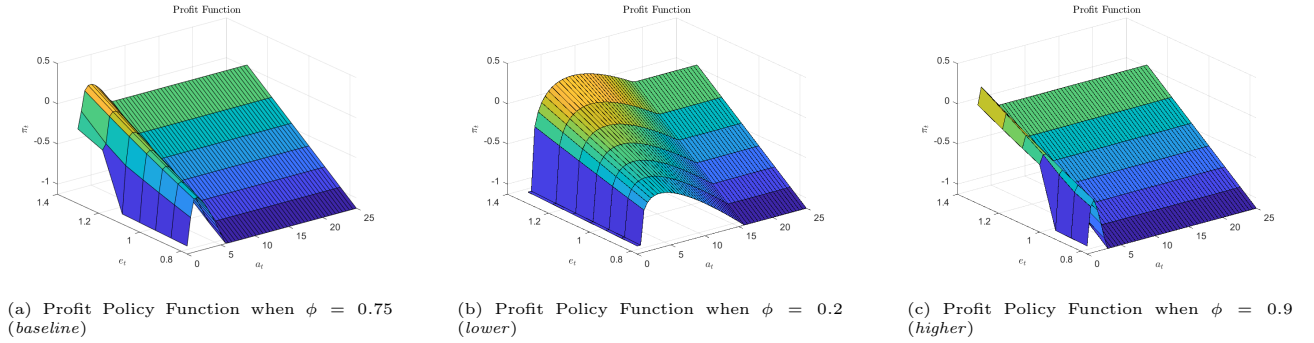


Figure 27: Profit Policy Function for different ϕ

In the baseline scenario (fig. 27a, $\phi = 0.75$), profits are positive in high-ability regions, with a clear boundary between viable and non-viable entrepreneurs. In figure 27b, $\phi = 0.2$, surprisingly, some profit levels appear higher despite tighter capital constraints. This results from looser enforcement: entrepreneurs face weaker consequences for default (as per equation 35), which allows them to take riskier, potentially more profitable positions—even if unsustainable. This creates non-monotonicities in the profit surface, where some mid-ability entrepreneurs exploit the enforcement gap. In contrast, figure 27c, $\phi = 0.9$ shows a more disciplined profit surface. Although stronger enforcement permits more capital rental overall (seen in the rental limit figure), the requirement to fully repay limits profitability to truly productive individuals, reducing gains in marginal cases.

This outcome directly links to the static firm problem. Equation (34) determines profits from production minus costs, while the constraint $K_t \leq \bar{K}(a, \varepsilon, \phi)$ and the enforcement rule in equation (35) determine how much capital can be used. A low ϕ shrinks the punishment for default, distorting incentives and reducing feasible capital levels. A high ϕ increases the feasible capital region but demands stricter repayment, which limits opportunistic behavior and shifts entrepreneurship toward higher-productivity agents. These dynamics highlight how the institutional environment—especially legal contract enforcement—can powerfully shape entrepreneurial behavior and aggregate economic outcomes.

Consumption of Self-Employed Similar to section 5.3, entrepreneurs' optimal consumption is also highly sensitive towards the change of profit as ϕ changes, as shown in Figure 28.

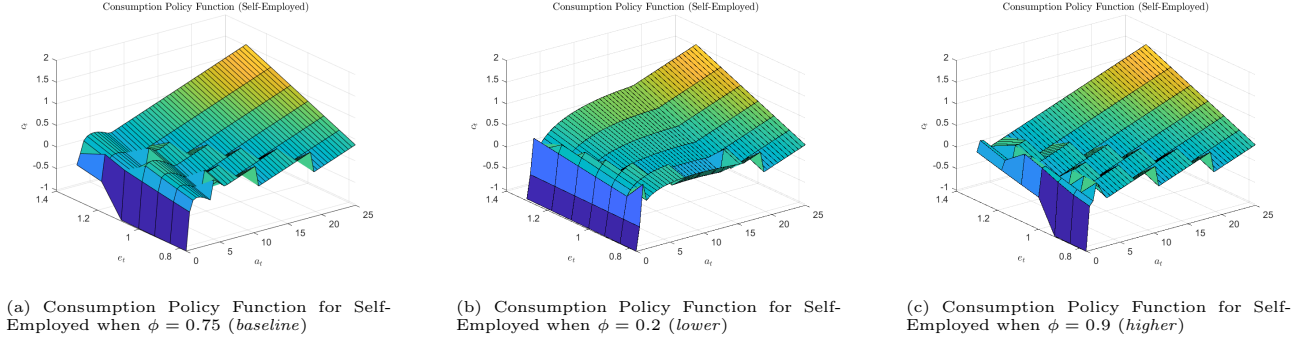


Figure 28: Consumption Policy Function for workers for different ϕ

In the baseline case (fig. 28a, $\phi = 0.75$), consumption rises with ability, and clear thresholds delineate where entrepreneurship becomes viable. This reflects moderate access to capital and enforcement, allowing reasonably productive entrepreneurs to consume more while managing savings.

In the weak enforcement scenario (fig. 28b, $\phi = 0.2$), despite tight capital constraints, consumption does not decline as sharply as expected. This mirrors earlier findings on profits—strategic default opportunities emerge, enabling some entrepreneurs to preserve or even boost consumption temporarily. However, the consumption surface becomes more irregular, reflecting instability and greater risk. In contrast, under strong enforcement (fig. 28c, $\phi = 0.9$), consumption surfaces remain largely similar to the baseline, with only modest gains for high-ability individuals. Although these entrepreneurs can rent more capital, strict repayment obligations temper consumption increases.

These results emphasize the non-linear welfare effects of ϕ . Stronger enforcement improves credit allocation but does not always lead to higher consumption due to repayment burdens. Meanwhile, weaker enforcement allows certain opportunistic behaviors, softening consumption losses in constrained settings. The link between profits and consumption via the intertemporal budget constraint ($c_t + a_{t+1} = (1+r)a_t + \pi_t$) means entrepreneurs dynamically adjust consumption and savings in response to institutional frictions.

6. Contract Enforcement on Policy Functions in General Equilibrium

In a stationary competitive equilibrium where the interest rate r and the wage rate w adjust to clear the capital and labor markets, contract enforcement affects the policy functions through its influence on borrowing constraints (the rental limits $\bar{k}^j(a, z_j, \phi)$ (see Section 2.1.4), which in turn influence agents' decisions.

Revisiting section 2.1.4, contract enforcement determines how strictly financial intermediaries limit the amount of capital an agent can rent relative to their own wealth. With perfect enforcement, intermediaries are willing to rent out more capital relative to an agent's wealth (i.e., more generous $\bar{k}^j(a, z_j, \phi)$), because they can recover their investment if the borrower defaults. Vice versa, with imperfect enforcement, rental limits are tighter (i.e., $\bar{k}^j(a, z_j, \phi)$ is smaller), reducing the agent's effective capital choice set.

Contract enforcement significantly shapes the individual policy functions in the economy, beginning with capital demand $k(a, z)$.

Capital Demand $k(a, z)$ Contract enforcement has a direct impact on the capital demand of entrepreneurs. With better enforcement, borrowing constraints are relaxed, allowing agents—especially

those with high productivity z_j but low wealth a —to access more capital through financial intermediaries. This leads to a higher optimal capital demand $k(a, z)$, particularly in capital-intensive occupations such as manufacturing. In contrast, worse enforcement tightens borrowing constraints, leading to lower $k(a, z)$ and discouraging entry into occupations that require substantial capital.

Occupational Choice $o(a, z)$ The occupational decision is also shaped by enforcement quality. Agents with strong entrepreneurial ideas (high z) but limited wealth (low a) may opt to become wage workers under poor enforcement regimes because they cannot rent sufficient capital to make entrepreneurship viable. When enforcement improves, these agents can better finance productive projects, allowing them to shift from wage employment to self-employment or entrepreneurial occupations, thereby making fuller use of their talents.

Labor Demand $l(a, z)$ Since labor is complementary to capital in production functions, the ability to rent more capital due to better enforcement also raises the optimal labor demand $l(a, z)$. Entrepreneurs can now afford to hire more labor, increasing employment and influencing the aggregate labor market. This upward pressure on labor demand can affect the equilibrium wage w , introducing general equilibrium feedback effects into individual labor demand decisions.

Consumption $c_S(a, z), c_M(a, z)$ Higher enforcement quality enhances production efficiency and profitability, enabling entrepreneurs to consume more. Specifically, both services consumption $c_S(a, z)$ and manufactured goods consumption $c_M(a, z)$ can rise when agents earn higher profits due to increased capital and labor usage. On the other hand, poor enforcement restricts capital access, lowers entrepreneurial profits, and consequently reduces consumption levels for constrained agents.

Savings $a'(a, z)$ Enforcement conditions also influence savings behavior. Under tighter constraints, agents tend to increase their precautionary savings to accumulate wealth for future investment, as they rely more on self-financing. This shifts the wealth distribution $G(a, z)$ toward higher individual saving rates, particularly among those anticipating future entrepreneurial activity. With better enforcement, the need for precautionary saving diminishes, as external capital becomes more accessible, potentially lowering overall savings rates and reshaping the stationary distribution of wealth and ideas.

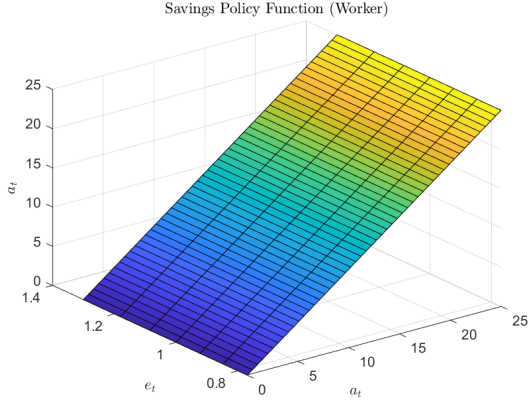
Overall, when the economy relaxes borrowing constraints, it allows more agents, especially those with high entrepreneurial ability but low initial wealth, to rent capital and enter productive occupations. This surge in capital demand places upward pressure on the rental rate r . Simultaneously, as more agents become entrepreneurs, the demand for labor rises since labor is a complement to capital in production. This increased labor demand pushes wages w upward. These equilibrium price changes, in turn, influence individual decision-making through their effects on the policy functions. A higher rental rate r increases the return to savings, which may modify agents' saving behavior. Similarly, a higher wage w alters the relative attractiveness of wage employment versus entrepreneurship, reshaping occupational choice decisions.

Enhanced enforcement also impacts the long-run distribution of wealth and entrepreneurial ability $G(a, z)$. With improved access to capital markets, more individuals with high productivity levels are able to become entrepreneurs and accumulate wealth more rapidly. Over time, this leads to an expansion in the mass of wealthier agents, reducing the number of high-productivity individuals trapped in low-capital, low-income occupations. As a result, the wealth distribution becomes less skewed, reflecting greater mobility and more efficient allocation of resources across occupations.

Appendices

Appendix A Baseline Policy Functions

A.1 Saving policy functions



(a) Savings Policy Function (Worker)



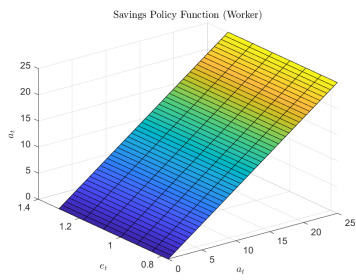
(b) Savings Policy Function (Self-Employed)

Figure 29: Savings Policy Function of (29a) Worker, and (29b) Self-Employed Individuals

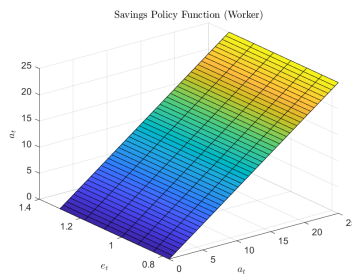
Appendix B Sensitivity of Policy Functions when changing technological parameters

B.1 Other Policy functions at different values of real wages

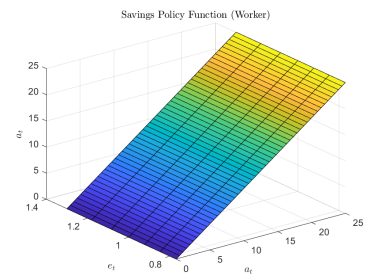
1. Saving Policy Functions for Workers and Self-Employed



(a) Savings Policy Functions of Workers when $w = 0.75$ (baseline)



(b) Savings Policy Functions of Workers when $w = 0.85$ (higher)



(c) Savings Policy Functions of Workers when $w = 1.0$ (higher)

Figure 30: Savings Policy Functions of Workers for different wage rates

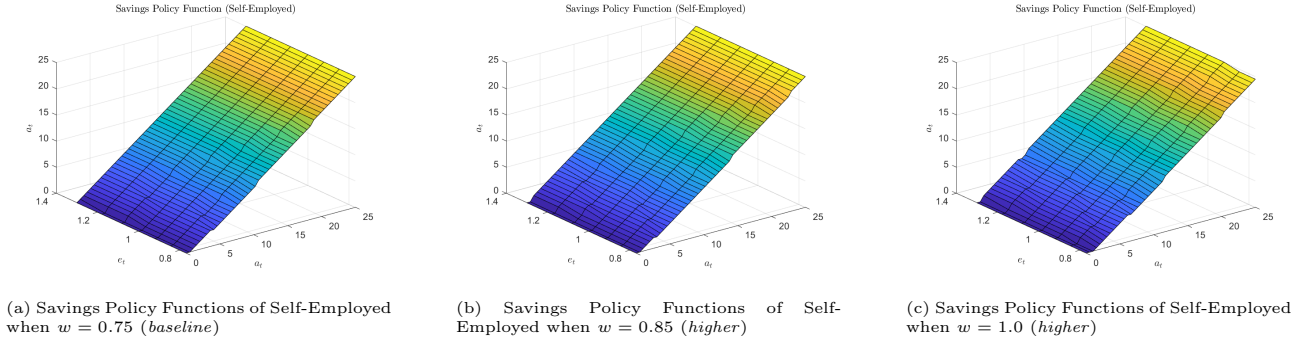


Figure 31: Savings Policy Functions of Self-Employed for different wage rates

2. Capital Rental Limit

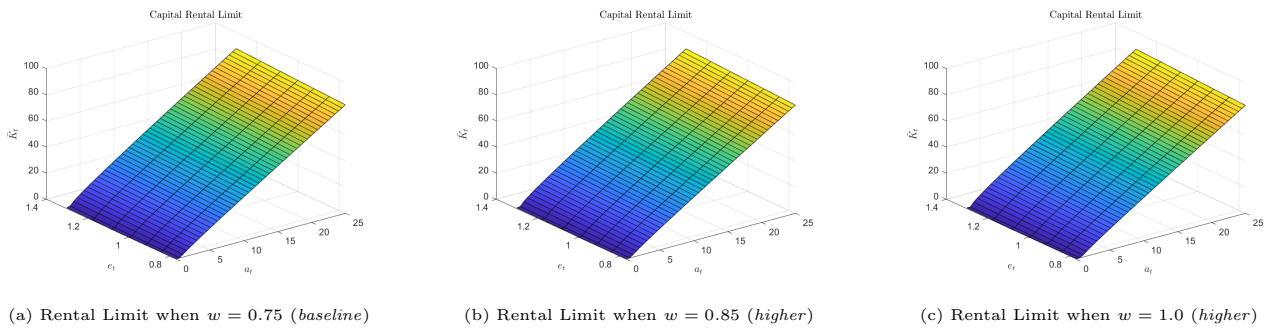


Figure 32: Rental Limit of Workers for different wage rates

B.2 Other Policy functions at different values of depreciation rates

1. Consumption Policy Function of Workers

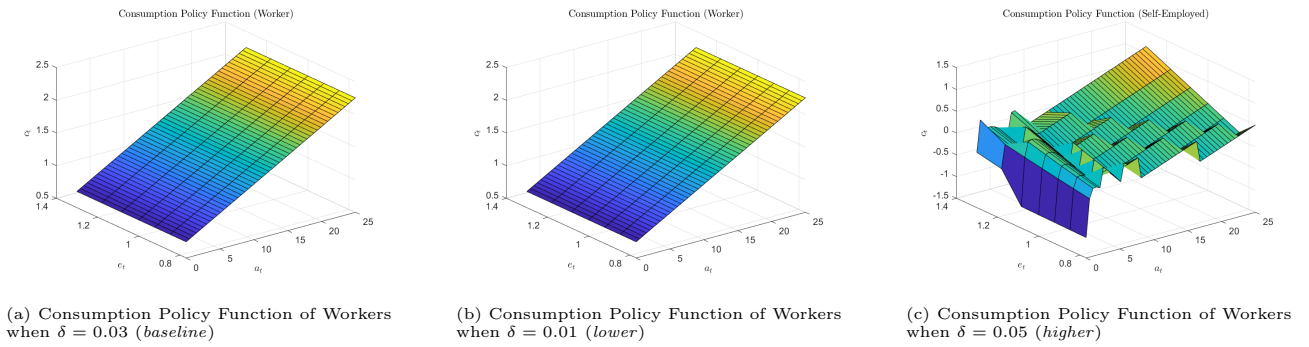


Figure 33: Consumption Policy Function of Workers for different δ

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